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Summary

Reference Number	NIHR304651
Lead Applicant	Mr Danyal Khan
Research Title	Towards AI-Assisted Pituitary Surgery – The Systematic Translation of Surgical Workflow Analysis to Improve Training, Audit and Prognostication.
Plain English Summary	Pituitary adenomas are amongst the most common brain tumours, and can cause vision problems and hormonal imbalances. They can lead to severe health consequences and even death if not treated promptly. Surgery through the nasal sinuses (transsphenoidal surgery) is usually the best treatment, but it's complex and requires years of surgical training. Different surgeons have different approaches, making it hard to track their performance and predict patient outcomes. This limits post-operative care options, like identifying low-risk patients for faster discharge. To begin to address these challenges we have to be able to understand and analyse the difference between surgeries, and we can do this using a process called workflow analysis. Workflow analysis breaks down each surgery into its basic building blocks (i.e. the surgical steps), and can be done by reviewing and annotating surgical videos. These labels can help us scrutinise and compare each step of the surgery, and for example, help us identify good performance. However, performing workflow analysis on every surgery performed takes a huge amount of time and resources, and would be very difficult to do consistently over time. Artificial intelligence (AI) can review videos and perform this analysis automatically, accurately, and quickly. This has only been possible with recent advances in AI, which continues to rapidly develop. Using predoctoral fellowship funding, we have built a secure and ethically approved system for collecting surgical videos and data at our centre, the National Hospital for Neurology and Neurosurgery. We have used this data to build AI systems capable of automatic workflow analysis in pituitary surgery. In this proposal, I plan to develop the next stage of this technology by developing three AI-powered clinical tools: 1) **Smart Operative Notes:** using AI workflow analysis to help surgeons create accurate, complete, and standardized reports – easing administrative burdens and improving tracking of performance. 2) **Real-Time Surgical Coach:** providing surgeons with AI guidance during and after surgery – highlighting upcoming steps, necessary tools, and even performance metrics. 3) **Personalized outcome prediction:** using AI to predict outcomes after workflow analysis of each surgery – to enable post-operative care plans tailored to each individual patient. I plan to develop these tools over 3 years. Each tool will undergo a pilot study to assess whether it's feasible and user-friendly. This will be followed by a larger study to explore whether the tool is safe, and whether it's effective (e.g. improves surgical outcomes). All of this will be performed at the UK's largest pituitary centre (National Hospital for Neurology and Neurosurgery, London), collaborating with engineers based at a prestigious surgical innovation institute (WEISS, University College London), and some of the world's largest medical AI companies (Medtronic and NVIDIA). This technology has the potential to revolutionise pituitary surgery by enhancing training, maintaining standards and facilitating personalised patient care - aligning with the NIHR Brain Tumours Highlight Notice. Similarly, If clinical translation is successful, these applications could be explored for other surgeries. The potential for this work is therefore vast, dovetailing with the NIHR's Health Data Science strategic theme and the NHS's National Strategy for AI. I believe this proposal has the ingredients necessary to actualise these potential benefits – a diverse team, based at centres of excellence, researching the fastest-growing technology in history, to improve patient outcomes.

Our purpose for collecting this data is to communicate with you about your application and have the necessary information to evaluate you for a grant. The data we collect here is collected in the public interest. Information provided here may be subject to Freedom of Information requests.

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1. Application Summary Information

Host organisation (which will administer any award)
University College London

Partner organisation (if applicable)	

Application Type

Strategic Theme	Health data science

Partnership Fellowship	N/A

WTE % Please select which WTE option you would prefer
1.00

Duration Doctoral Fellowship is a 3 year award and based on your WTE preference the duration in months is calculated as follows
36

Research Title	Towards AI-Assisted Pituitary Surgery – The Systematic Translation of Surgical Workflow Analysis to Improve Training, Audit and Prognostication.

Research Type	Feasibility study, Pilot study

Proposed start date if grant awarded	01/02/2025

2. Applicant CV

Degrees and Professional Qualifications

From	To	Qualification	Subject	Country	Institution	Class	Department
08/2019	08/2020	AFHEA	Associate Fellowship of the Higher Education Academy	United Kingdom	Higher Education Academy	N/A	N/A
09/2018	05/2019	MRCS	Completed MRCS part A&B during FY1 (87% across both parts)	United Kingdom	Royal College of Surgeons of England	N/A	N/A
08/2016	04/2022	ECFMG certification (completion of USMLE exams)	USMLE Step 1 08/2016 (264/300) USMLE Step 2CS 10/2017 (Pass) USMLE Step 2 CK 04/2022 (239/300)	United States	ECFMG	N/A	N/A
09/2013	05/2018	MBBS	Medicine (Undergraduate)	Ireland	Royal College of Surgeons in Ireland	2018	School of Medicine

Present and previous positions

(including from/to dates, job title and employer)

From	To	Position	Organisation	WTE % Research
08/2021	-	NIHR Academic Clinical Fellow in Neurosurgery and Neurosurgical Trainee (run-through)	University College London / London Deanery	25
08/2020	08/2021	WEISS Clinical Research Fellowship in Artificial Intelligence, Robotics and Neurosurgery	University College London	50
07/2018	08/2020	Academic Foundation Programme Trainee in Clinical Neuroscience	University of Cambridge	17

Research Grants held

Title of Grant	Role in the Research Grant	Total Grant £	Source of Grant (Organisation Name)	Duration (In Months)	Start Date of Funding	End Date of Funding
Pre-Doctoral Fellowship (Cancer Research UK)	I was the sole fellowship recipient for my project "Personalised, data-driven surgical management of pituitary tumours using artificial intelligence – a pilot study". Nationally competitive award.	24992	Cancer Research UK	12	01/03/2022	01/03/2023
Berkeley Travelling Fellowship	Sole fellowship recipient for the project "Artificial Intelligence Driven Analysis of Surgical Gestures during Pituitary Adenoma Surgery - Framework Generation and Pre-Clinical Development" which I will be completing at Stanford	8125	University College London & University of Cambridg	2	26/02/2024	26/04/2024

Title of Grant	Role in the Research Grant	Total Grant £	Source of Grant (Organisation Name)	Duration (In Months)	Start Date of Funding	End Date of Funding
	University, USA. Nationally competitive award open to all medical graduates in the UK.					
Small Acorns Grant	Co-investigator for a locally competitive project grant: “Research priority setting for pituitary adenomas”	2700	National Brain Appeal	12	01/09/2023	01/09/2024
WEISS Public Engagement Award	Co-investigator for a locally competitive project grant: “Research priority setting for pituitary adenomas”	2700	WEISS, University College London	12	01/09/2023	01/09/2024
Stryker Collaborative Research Grant	Co-investigator on an internationally competitive project grant from industry. Awarded to execute a multicentre collaborative research project "Quality of Life after Endonasal Skull Base Surgery for Pituitary Lesions - A Multicentre Prospective Study". This was awarded based on my track record executing such studies within the field.	50000	Stryker	24	01/12/2023	01/12/2025

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(2024) ‘Starup-Hansen J, Williams SC, Valetopoulou A, Khan DZ, Horsfall HL, Moudgil-Joshi J, Burton O, Kanona H, Saeed SR, Muirhead W, Marcus HJ. Skull Base Repair following Resection of Vestibular Schwannoma: A Systematic Review (Part 1: The Retrosigmoid Approach). Journal of Neurological Surgery Part B: Skull Base.’,
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Marcus, HJ., Ramirez, PT., Khan, DZ., Layard Horsfall, H., Hanrahan, JG., Williams, SC., Beard, DJ., Bhat, R., Catchpole, K., Cook, A., Hutchison, K., Martin, J., Melvin, T., Stoyanov, D., Rovers, M., Raison, N., Dasgupta, P., Noonan, D., Stocken, D., Sturt, G., Vanhoestenberghe, A., Vasey, B. and McCulloch, P.. (2024) ‘The IDEAL framework for surgical robotics: development, comparative evaluation and long-term monitoring.’, *Nature medicine*, 30(1).
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Enkaoua, A., Islam, M., Ramalhinho, J., Dowrick, T., Booker, J., Khan, DZ., Marcus, HJ. and Clarkson, MJ.. (2023)

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Das, A., Bano, S., Vasconcelos, F., Khan, DZ., Marcus, HJ. and Stoyanov, D.. (2022) ‘Reducing prediction volatility in the surgical workflow recognition of endoscopic pituitary surgery.’, *International journal of computer assisted radiology and surgery*, 17(8).
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Tetreault, L., Mowforth, O., Khan, DZ., Gronlund, T., Garwood, P., Hazenbiller, O., Harrop, JS., Aarabi, B., Rahimi-Movaghar, V., Kurpad, SN., Guest, JD., Wilson, JR., Kwon, BK., Fehlings, MG., Davies, BM. and Kotter, MRN.. (2022) ‘James Lind Alliance Priority Setting Partnership for Degenerative Cervical Myelopathy [AO Spine RECODE-DCM]: An Overview of the Methodology Used to Process and Short-List Research Uncertainties.’, *Global spine journal*, 12(1_suppl).
Koh, CH., Khan, DZ., Digpal, R., Layard Horsfall, H., Ali, AMS., Baldeweg, SE., Bouloux, PM., Dorward, NL., Drake, WM., Evanson, J., Grieve, J., Stoyanov, D., Korbonits, M. and Marcus, HJ.. (2022) ‘The clinical outcomes of imaging modalities for surgical management Cushing's disease - A systematic review and meta-analysis.’, *Frontiers in endocrinology*, 13.
(2022) ‘CSF rhinorrhoea after endonasal intervention to the skull base (CRANIAL): A multicentre prospective observational study.’, *Frontiers in oncology*, 12.

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Mowforth, OD., Khan, DZ., Wong, MY., Pickering, GAE., Dean, L., Magee, J., Mullarkey, L., Hirayama, Y., Rihova, M., Butler, M., Stewart, M., Goulson, B., Ahmed, S., Fricke, K., Popa-Nimigean, V., Millar, Z., Venkatesh, A., Willison, A., Senthil, K., Hazenbiller, O., Sarewitz, E., Sadler, I., Gronlund, T., Tetreault, L., Harrop, JS., Aarabi, B., Rahimi-Movaghar, V., Kurpad, SN., Guest, JD., Wilson, JR., Kwon, BK., Fehlings, MG., McNair, AGK., Davies, BM. and Kotter, MRN.. (2021) 'Gathering Global Perspectives to Establish the Research Priorities and Minimum Data Sets for Degenerative Cervical Myelopathy: Sampling Strategy of the First Round Consensus Surveys of AO Spine RECODE-DCM.', *Global spine journal*, 12(1_suppl).
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Publication Record

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Relevant Prizes, Awards and other Academic Distinctions

Please note the information for additional prizes or awards is automatically populated from information provided in the "Update CV" section of the "Manage My Details" page, which you can update by following this link https://aramis.nihr.ac.uk/MyAccount/UpdateCV.aspx

Date (mm/yyyy)	Source	Prize	Role of Applicant
01/12/2023	Society of British Neurological Surgeons	UK Trainee Conference Bursary Award. Nationally competitive award (£500) based on academic merit, which supported attendance at the World Federation of Neurosurgical Societies annual congress.	Lead applicant & sole recipient
01/09/2022	Society of British Neurological Surgeons	Sir Hugh Cairns Prize. Nationally competitive award (£1000) based on my research: "CSF rhinorrhoea	Lead applicant & sole recipient

Relevant Prizes, Awards and other Academic Distinctions

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Date (mm/yyyy)	Source	Prize	Role of Applicant
		after endonasal intervention to the skull base - a multicentre prospective observational study"	
01/09/2022	Faculty Opinions.org	Academic Travel Award ($500), awarded for critical appraisal skills in producing online research summaries. Awarded on merit, in international competition with other writers for the organisation.	Lead applicant & sole recipient
01/05/2022	Queen Square Neurosurgery	Research and Audit Prize (£500), awarded locally to the neurosurgical trainee with the best academic performance for the calendar year.	Sole recipient
01/05/2022	Queen Square Neurosurgery	Microsurgical Skills Trainee Award (£500), awarded locally to the neurosurgical trainee with the best performance in the annual microsurgical skills simulation competition.	Sole recipient
01/04/2021	Society of British Neurological Surgeons	Barrow Foundation UK Junior Neurosurgical Trainee Award. Awarded for ranking 1st in National Neurosurgical Selection. Provides £3000 funding to visit the Barrow Neurological Institute (Arizona, USA).	Sole recipient
01/11/2020	University of Cambridge, Clinical Medicine Society	Clinical Teaching Excellence Award. Awarded to the top-ranked university Clinical Supervisors as rated by university students on their clinical rotations.	Sole recipient
01/05/2018	Royal College of Surgeons in Ireland	Numerous medical school prizes: -2nd Place in Graduating Class Award (of >350 students) -Sheppard Prize in Pathology (1st of >350 students) -Quinn Medal in Ophthalmology (1st of >350 students)	Sole recipient
01/11/2017	Royal College of Surgeons in Ireland	Collaborative Elective Scholarship ($2000, locally competitive) awarded based on academic merit to attend a 1-month long clinical sub-internship at Johns Hopkins	Lead applicant & sole recipient

Relevant Prizes, Awards and other Academic Distinctions

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Date (mm/yyyy)	Source	Prize	Role of Applicant
		(Baltimore, USA).	
01/11/2017	National University of Ireland	HH Stewart Medical Prize in Biochemistry (2nd place). Awarded for my performance in a national written exam between the highest-ranking students across multiple Irish medical schools.	Sole recipient
01/08/2017	Mayo Clinic Rochester	Sub-Internship Bursary ($3000, internationally competitive) awarded to attend Mayo Clinic for a 1-month long clinical sub-internship whilst a final year medical student.	Lead applicant & sole recipient
01/06/2017	Royal College of Surgeons in Ireland	Clinical Elective Grant (€500, locally competitive) awarded based on academic merit to attend Emory University (Atlanta, USA) for a 1-month long clinical sub-internship.	Lead applicant & sole recipient
01/11/2015	National University of Ireland	HH Stewart Medical Prize in Physiology (2nd place). Awarded for my performance in a national written exam between the highest-ranking students across multiple Irish medical schools.	Sole recipient
01/11/2011	The European Union Olympiads	Irish European Union Junior Olympiads national finalist in Science & Mathematics	Sole recipient

ORCID iD 0000-0001-9213-2550

3. Applicant Research Background

Professional Background	Medically Qualified

Specialty (if Medically Qualified)
Neurosurgery

Do you have a GMC number?	Yes

GMC number Please supply a valid GMC number	7647204

Please describe your research career to date

I aim to be an academic neurosurgeon interested in applying technology and innovation to address neurosurgical problems. In pursuit of this, I have completed formal clinical academic training programmes throughout my postgraduate career. I am now a run-through neurosurgical trainee (London Deanery) and an NIHR Academic Clinical Fellow at University College London (UCL), after ranking first in Neurosurgery National Selection and attaining the Barrow Foundation Award from the Society of British Neurosurgeons (SBNS). During this time, I have published 70 peer-reviewed publications (including within Nature Medicine), and attained multiple fellowships and grants. Completing a PhD is the next step in my development and training as a surgeon-scientist.

Through needs analysis, targeted innovation, and systematic evaluation I have gained a skill set in surgical innovation. This began as a visiting medical student during a research elective at the Hamlyn Centre, Imperial College London in 2016. I conducted a literature review of unmet clinical needs, followed by a technological gap analysis, using this to collaborate with engineers and conceive a robotic platform which used a parallel tendon manipulator within a tubular retractor system. We then generated a prototype, and executed a comparative preclinical validation study, presenting the findings internationally via a clinical and engineering conference. This project was my first exposure to the principles of biological design and interdisciplinary collaboration, and it allowed me to discover a creative outlet which I have exercised since. My supervisor, Mr Hani Marcus, has remained my mentor and role model since (8 years later).

I developed these skills further during my Academic Foundation Programme (Clinical Neurosciences) at the University of Cambridge. I applied novel robotic transcranial Doppler technology with finger plethysmography to noninvasively assess cerebral autoregulation in patients with post-concussive syndrome. I established an unmet need through a literature review, and translated this technology via a first-in-human pilot study - after attaining ethical approvals and collaborating with brain physicists.

I then joined WEISS (Wellcome/EPSRC Centre for Interventional and Surgical Sciences) UCL as a clinical research fellow in AI and neurosurgery, where I developed an AI model capable of workflow recognition for endoscopic pituitary surgery in collaboration with engineers and industry. This was the first of its kind in pituitary surgery, and I then attained an NIHR Academic Clinical Fellowship and Cancer Research UK Predoctoral Fellowship (£25000) to develop these AI workflow models further, publishing numerous peer-reviewed articles, for example. I have also delivered multiple podium presentations on this work, including internationally at the World Federation of Neurosurgeons Congress, supported by a nationally competitive SBNS Trainee Travelling Bursary. This year, I will be a visiting scholar at Stanford University for 2 months after attaining a Berkeley Travelling Fellowship (£8000, UCL & University of Cambridge), where I will externally validate our AI models and further refine their workflow analysis performance.

Further to my skills in surgical technology, I have complementary skills in collaborative clinical research. I was a junior principal investigator for a multicentre observational pituitary surgery outcomes study, recruiting almost every neurosurgical centre in the UK & Ireland. I executed this collaborative study while I held a national leadership position (Vice Chair of the Neurology and Neurosurgery Interest Group), and leveraged this to motivate, coordinate and collaborate with a network of >100 consultants, trainees and students. This study recruited >800 patients, representing the largest study of its kind, leading to multiple publications and presentations (for example). For this, I was awarded the prestigious SBNS Hugh Cairns Prize, and have received a research grant from Stryker

(£50000, Co-investigator) for the next stage of this study (national patient-reported outcomes). This work has also created an effective pituitary research network, which I continue to manage and remains productive. I have applied these skills in teamworking across numerous other international projects, including with The Pituitary Society, therefore establishing myself within relevant national and international pituitary research networks, which I hope to leverage for future multicentre translation of my surgical technology work.

Additionally, I recognise the importance of community engagement as an academic and as a clinician. In terms of public engagement, I have showcased our AI work on the BBC, Reuters, NowThis and RadioFrance. I have also created numerous explanatory videos of my work via the UCL Surge and PitCOP projects. For the latter, I have received funding via the Small Acorns Fund (National Brain Appeal, £2700, Co-investigator) Public Engagement Grant (WEISS, £2700, Co-investigator) to investigate patient priorities in pituitary adenoma research and develop a core outcome set via consensus, as part of a research translation toolkit for my future work. In terms of research community engagement, I co-created weekly academic "tweetorials" of contemporary pituitary research via the British Neuroendoscopy Society Twitter for two years. I have served as a faculty member of the UCL AI in Neurosurgery Course, where I taught about critical appraisal of clinical AI research to an international audience, receiving excellent feedback. I also am passionate about open science, and have curated 25 open-access annotated pituitary surgery videos as part of the International Endoscopic Vision Challenge.

Finally, I received formal teaching training via the RCS Teaching the Trainers Course, and have attained an Associate Fellowship from the Higher Education Academy as an undergraduate Clinical Supervisor within the Integrated Foundation in Medical Education programme (University of Cambridge). I have applied these skills through numerous avenues since. For example, as a Trainee Lead at the Queen Square Neurosurgery Simulation Centre, where I have organised consecutive pituitary surgery simulation courses (receiving outstanding feedback, and demand for a 4th edition of the course) and have co-led simulation research. I am also the Skull Base Trainee Lead of the Queen Square Annual Review Course, and have led the coordination of pituitary surgery teaching sessions – attracting excellent feedback from international delegates for 3 consecutive years.

I have developed these skills in surgical innovation, collaborative research and community engagement whilst delivering on any funding I have been fortunate to receive. I believe I have the necessary skill set to maximise the opportunities for personal improvement and project development offered by this prestigious fellowship.

Have you already registered for a PhD?	No

Has this application, or a similar application, previously been submitted to this or any other funding body?
Yes

If Yes, provide details
Yes, a similar application for doctoral funding has been submitted to the Medical Research Council for consideration for a Pre-doctoral Clinical Research Training Fellowship. This was submitted in September 2023, but was rejected as it was considered out of remit. They concluded that the bulk of the application was within the remit of the NIHR, rather than the MRC.

Contextual Factors
Please use this question to detail any contextual factors you wish to make the Selection Committee aware of. NIHR wants to know about any circumstances so that they may take them into consideration during the assessment of your application. Contextual factors may include:

- Career breaks due to parental leave, or periods of illness.
- Reduced time spent undertaking research due to a disability or caring responsibilities. This could include any physical or mental difficulty that may have impacted your research career. These are situations that have a significant impact on your ability to undertake research.
- Reduced opportunities to career support e.g. mentorship, and limited opportunities to undertake prior research and training.
- Impact of the COVID-19 pandemic on your research career.

Please also use this section to detail any other factors that may have impacted your research career not listed in the examples provided. The impact on your career to date will be specific to your particular circumstances but could include such impacts as limited opportunities to obtain grant funding, or fewer publications. In general terms, mitigating factors should be significant, and relevant.

NIHR acknowledges that you may be reluctant, or uncomfortable disclosing relevant information that is sensitive. However, you should bear in mind that we are unable to take into account factors that you do not disclose. Please be assured that information provided by you is sensitive and will be treated confidentially and in line with General Data and Protection Regulations (GDPR).
Nothing insurmountable, for which I am grateful.

4. Plain English Summary of Research

Pituitary adenomas are amongst the most common brain tumours, and can cause vision problems and hormonal imbalances. They can lead to severe health consequences and even death if not treated promptly. Surgery through the nasal sinuses (transsphenoidal surgery) is usually the best treatment, but it's complex and requires years of surgical training. Different surgeons have different approaches, making it hard to track their performance and predict patient outcomes. This limits post-operative care options, like identifying low-risk patients for faster discharge.

To begin to address these challenges we have to be able to understand and analyse the difference between surgeries, and we can do this using a process called workflow analysis. Workflow analysis breaks down each surgery into its basic building blocks (i.e. the surgical steps), and can be done by reviewing and annotating surgical videos. These labels can help us scrutinise and compare each step of the surgery, and for example, help us identify good performance. However, performing workflow analysis on every surgery performed takes a huge amount of time and resources, and would be very difficult to do consistently over time. Artificial intelligence (AI) can review videos and perform this analysis automatically, accurately, and quickly. This has only been possible with recent advances in AI, which continues to rapidly develop.

Using predoctoral fellowship funding, we have built a secure and ethically approved system for collecting surgical videos and data at our centre, the National Hospital for Neurology and Neurosurgery. We have used this data to build AI systems capable of automatic workflow analysis in pituitary surgery. In this proposal, I plan to develop the next stage of this technology by developing three AI-powered clinical tools:

1) **Smart Operative Notes:** using AI workflow analysis to help surgeons create accurate, complete, and standardized reports – easing administrative burdens and improving tracking of performance.
2) **Real-Time Surgical Coach:** providing surgeons with AI guidance during and after surgery – highlighting upcoming steps, necessary tools, and even performance metrics.
3) **Personalized outcome prediction:** using AI to predict outcomes after workflow analysis of each surgery – to enable post-operative care plans tailored to each individual patient.

I plan to develop these tools over 3 years. Each tool will undergo a pilot study to assess whether it's feasible and user-friendly. This will be followed by a larger study to explore whether the tool is safe, and whether it's effective (e.g. improves surgical outcomes). All of this will be performed at the UK's largest pituitary centre (National Hospital for Neurology and Neurosurgery, London), collaborating with engineers based at a prestigious surgical innovation institute (WEISS, University College London), and some of the world's largest medical AI companies (Medtronic and NVIDIA).

This technology has the potential to revolutionise pituitary surgery by enhancing training, maintaining standards and facilitating personalised patient care - aligning with the NIHR Brain Tumours Highlight Notice. Similarly, If clinical translation is successful, these applications could be explored for other surgeries. The potential for this work is therefore vast, dovetailing with the NIHR's Health Data Science strategic theme and the NHS's National Strategy for AI.

I believe this proposal has the ingredients necessary to actualise these potential benefits – a diverse team, based at centres of excellence, researching the fastest-growing technology in history, to improve patient outcomes.

5. Scientific Abstract

This proposal leverages artificial intelligence (AI) to address key unmet surgical needs, with the ultimate goal of improving patient outcomes, using pituitary surgery as an exemplar.

Pituitary tumours are amongst the most common brain tumours, presenting in a variety of ways, including visual decline and hormonal imbalance. Left untreated, they may result in significant morbidity, mortality and quality of life impairment. Transsphenoidal surgery is the treatment of choice for most patients with symptomatic adenomas. However, this is a challenging operation to perform, requiring years of dedicated training. Different surgeons perform these surgeries in distinct ways, with variance in the recording of surgical performance and resultant outcomes. Therefore, accurate auditing of practice is challenging. The variation in surgical performance, along with patient differences, also makes the prediction of post-operative outcomes difficult. Consequently, it is challenging to tailor post-operative care to each patient, for example, recognising patients who may be at low risk for complications and safely discharged earlier.

The first step to tackling these problems is to understand how these surgeries are performed and compare the relationships between their performance and outcomes. Operative workflow analysis enables this by breaking down operations into the fundamental standardised building blocks, including the surgical steps performed. Practically, this analysis is performed by segmenting surgical videos. This is labour-intensive if undertaken manually but artificial intelligence (AI) has the potential to perform this analysis automatically, accurately, and at scale.

We have developed pre-clinical AI models capable of recognising surgical workflow in endoscopic pituitary surgery. The next step is to apply this technology to the unmet clinical needs of pituitary surgery, and I plan to do this over a 3-year doctoral fellowship, guided by the IDEAL and DECIDE-AI guidelines, with 3 aims:

1) To investigate the use of AI workflow analysis to **improve audit and quality improvement** by assisting surgeons in creating semi-automated operative notes. We hypothesize these will be more accurate, comprehensive and standardised, whilst helping relieve administrative burden.
2) To examine the uses of AI workflow analysis to **address the training needs** of surgical teams, by presenting a real-time analysis of current and upcoming surgical steps and instruments, with surgical performance metrics. We hypothesize this will improve the educational yield from each case by orientating the team during the procedure and by facilitating coaching by mentors.
3) To explore the use of AI workflow analysis for **prognostication** by linking this analysis with clinical outcomes, allowing data-driven post-operative outcome prediction. We hypothesize this will allow more patient-specific post-operative management.

I plan to translate each of these technologies via pilot studies (examining feasibility, acceptability and usability) and single centre development studies (examining effectiveness and safety).

We have an established team of neurosurgeons (at the UK's largest pituitary centre), engineers (at a world-leading surgical innovation institute) and key industry collaborators (Medtronic and NVIDIA). Having built a secure data pipeline, generated a large dataset, and created foundational preclinical models through an NIHR Academic Clinical Fellowship and a Cancer Research UK Predoctoral Fellowship, I am confident I can deliver this proposal.

6. Detailed Research Plan

The overarching goal of this proposal is to clinically translate AI video analysis technology into practice, with the ultimate aim of improving outcomes for patients undergoing endoscopic pituitary surgery resection. This builds on work supported by my NIHR Academic Clinical Fellowship and CRUK Predoctoral Fellowship which have generated the largest known pituitary surgery video and outcomes dataset. The work will be carried out in the UK's highest volume pituitary surgery centre, supported by key industry partners and a world-leading surgical innovation institute which holds related project grants to provide the technical and engineering infrastructure required. I hope that at least one (if not all) of the three proposed work streams will be translated further in the future, via multicentre comparative studies.

1. What are the problems being addressed?
Pituitary adenomas are among the most common intracranial tumours, with an estimated prevalence of up to 20%[1, 2]. They may present incidentally, through mass effect (e.g. visual decline) or hormone imbalance (e.g. acromegaly), therefore potentially causing significant morbidity, quality of life reduction and death if left untreated[1, 3]. The gold standard treatment for most patients with symptomatic pituitary adenoma is transsphenoidal surgical excision, with the endoscopic transsphenoidal approach (eTSA) bolstering access to these tumours[4].

However, these operations are technically demanding with long learning curves, posing a training challenge for surgeons, and therefore requiring post-training fellowships to attain the necessary clinical and technical skills[5, 6]. Furthermore, there is variation in how these operations are performed (largely based on surgeon preference), which exacerbates this training challenge and may result in differing outcomes[3, 7]. Through executing a systematic review and a large multicentre observational study, I have found significant heterogeneity in the closure phase alone - in how it was performed, how operative details were recorded and how it affected outcomes (i.e. whether it prevented the leakage of cerebrospinal fluid [CSF])[8-12]. This heterogeneity in practice and operative documentation poses a significant challenge to the clinical audit of performances and maintaining safety. Similarly, the resulting wide variations in outcomes, dependent on both operative and patient factors, make post-operative prognostication notoriously difficult[12-14]. Therefore, patients are often monitored for 3-5 days to monitor for common complications (e.g. sodium imbalance), and initiatives to facilitate earlier discharge or prophylactic treatments (e.g. fluid restriction) in selected cases have not become widespread[12-14]. The inability to robustly predict outcomes and stratify patients poses a fundamental barrier to these initiatives[13, 14].

To begin to address these unmet audit, training and clinical effectiveness needs, we generated an international, consensus-driven analysis of the operative workflow (surgical steps, instruments and errors) in contemporary pituitary surgery[15-17]. This workflow analysis is used to systematically codify operations into their fundamental components, to allow in-depth analysis and identification of operative variations[15]. This can be performed through the segmentation of operative videos. However, done manually, this is extremely labour-intensive - posing limitations to its upscaling and analysis. Recent advances in artificial intelligence (AI), more specifically machine learning and computer vision, have allowed this operative workflow analysis to be generated automatically[18, 19]. Thus, we have developed novel pre-clinical AI models capable of accurately and automatically recognising pituitary surgery workflow from operative videos – a world first for pituitary surgery[20-23].

2. Importance of this research
In this proposal, pituitary surgery serves as an illustrative exemplar for exploring the benefits of this novel technology in surgery, aligning with the NIHR Brain Tumours Highlight Notice. However, our findings could inform improvements in the audit, training and outcomes of any type of surgery, once a data pipeline is established. The potential for this work is therefore vast, dovetailing with the NIHR's Health Data Science strategic theme and the NHS's National Strategy for AI.

3. This research in the context of existing literature
I have conducted multiple reviews exploring the above clinical problems and potential technological solutions such as AI. This includes a large 193 study systematic review, highlighting the considerable heterogeneity in the performance of endoscopic pituitary surgery, and resulting outcomes, using the closure phase as an exemplar[8]. I explored this further via a multicentre prospective observational study encompassing >700 cases and almost every pituitary surgery unit in the UK and Ireland, which echoed the findings of the review and highlighted the difficulty of predicting outcomes[11, 12]. When considering the reporting of baseline characteristics and outcomes more generally, another 178 study systematic review again revealed the heterogeneity in the selection, recording and analysis of this data within pituitary adenoma surgery[24, 25]. This dovetails with wider literature on major unmet training, audit and prognostication needs, which is rooted in this heterogeneity of practice and reporting.

Following this, I have conducted a narrative review of the potential of technological advances to address these needs in pituitary surgery, and the general role of AI within surgical theatres of the future[16, 26]. Numerous

systematic reviews specifically exploring the potential of AI-driven video workflow analysis have already been published, including Timoh et al's 34 study review highlighting the infancy and recent expansion of this field[27]. This is because the use of AI in this fashion was only recently possible, with the advent of neural networks and the computing power necessary to leverage them. Translating this technology for specific clinical applications (e.g. teaching, audit and decision support) is even more novel, so there are limited previous studies on this generally, and none of these are within pituitary surgery. Thus, this proposal seeks to generate new evidence in a rapidly developing space. Recognising the exploratory nature of this work, I have de-risked this proposal by creating three parallel project streams (as opposed to one).

4. Research aims

I aim to clinically translate the preclinical AI workflow analysis technology I have been developing – tailoring it to the unmet clinical needs of pituitary surgery[16]. I plan to do this over a 3-year doctoral fellowship, performing earlier clinical translation guided by the IDEAL and DECIDE-AI guidelines[28-31], with 3 aims:

1) To investigate the use of AI workflow analysis to **improve audit** by assisting surgeons in creating semi-automated operative notes[22]. We hypothesize these will be more accurate and comprehensive, whilst helping relieve some administrative burden.
2) To examine the uses of AI workflow analysis to **address the training needs** of surgical teams, by presenting a real-time analysis of current and upcoming surgical steps and instruments, with surgical performance metrics. We hypothesize this will improve the educational yield from each case by orientating the team during the procedure and by facilitating coaching by mentors[32].
3) To explore the use of AI workflow analysis for **prognostication** by linking this analysis with clinical outcomes, allowing data-driven post-operative outcome prediction. We hypothesize this will allow more patient-specific post-operative management (e.g. reducing the length of stay or improving complication recognition)[16].

5. Project plan

This work builds on foundations laid by my NIHR Academic Clinical Fellowship (2021-2024) and Cancer Research UK Predoctoral Fellowship (awarded March 2022, for 20,000gbp over 1 year). These programmes have allowed me to collect pilot data (annotated videos and associated clinical data) for >200 pituitary adenoma patients, the largest dataset of its kind. They have also supported the building of a pipeline for secure data collection (including regional ethics committee approval and industry collaboration), and the development of novel preclinical AI models capable of accurately and automatically recognising operative workflow[20-23].

My plan to systematically clinically translate this surgical video workflow AI technology is via three parallel work packages (WP) over three years (Figure 1, Research Timetable Figure). This plan is guided by the IDEAL framework for evaluating surgical innovation and DECIDE AI guidelines for early clinical studies of AI interventions, with the ultimate goal of readying this technology for subsequent multicentre comparative studies[28-31]. The chosen sample size for IDEAL Stage 1 studies (n=6) and IDEAL Stage 2a (n=30) is derived from the IDEAL Guidelines[31]. The intent of these studies is rapid iterative modification (of the AI-human interface), and preparation of the technology for future multicentre and definite comparative studies.

The WPs are listed in order of projected implementation difficulty (from easy to more challenging) and will be staggered as such (see Research Timetable Figure). WP1 could overlap with WP2 and WP3 if needed, but WP2 and WP3 will not overlap as they may confound each other's analysis. The chosen sample size for IDEAL Stage 1 studies (n=6) and IDEAL Stage 2a (n=30) is derived from the IDEAL Guidelines[31]. The intent of these studies is rapid iterative modification (of the AI-human interface), and preparation of the technology for future multicentre and definite comparative studies.

WP1: Applying AI workflow analysis to assist surgical audit.

The goal of this stage is to apply our pre-clinical AI model, which uses AI workflow analysis to recognise the steps performed during an operation and populate an operative note template, to improve the accuracy, efficiency and standardisation of our surgical documentation[22].

The next stage of this work is a first-in-human pilot study (IDEAL Stage 1) of the feasibility, acceptability and accuracy of AI-assisted operative notes, where the surgeon is presented with an AI-generated note immediately post-operatively (based on the operative video). This note will be copied and pasted into our EMR after the surgeon's editing to ensure representativeness and add context (i.e. information not possible to glean from video alone). Six prospective cases will used, recording the feasibility (number of AI-assisted notes completed, time taken) and acceptability (via structured questionnaire) as primary outcomes[28, 33]. Additionally, accuracy, comprehensiveness and clarity will be assessed via an expert surgeon of the video and operative note, using questionnaires based on our consensus framework for core surgical steps and the Royal College of Surgeon's guidance for operation documentation[15, 34]. Human factors analysis will be gathered via observation and interview, and used to iterate the AI output interface and usability[28]. The success criteria for progression to the next stage are based on these primary outcomes, if feasibility (>80% of notes completed successfully within 15 minutes) and

acceptability (>80% of participants finding the notes acceptable) are demonstrated. If the study fails on its first trial, the intervention will be iterated and re-trialled if deemed necessary.

We will then perform a single centre prospective development study (IDEAL Stage 2a) of 30 prospective cases, comparing accuracy, comprehensiveness and clarity to a retrospective cohort of 30 operative notes and videos with no AI documentation support. Again, the AI-human interface will be continually improved with iterative human factors analysis. The Data Collection section details outcome metrics in further detail. This study will provide the necessary experience and technological refinement for planning multicentre explorative (IDEAL Stage 2b) and comparative (IDEAL Stage 3) studies. We plan to pursue these (via a dedicated project grant in the future) if the intervention demonstrates any improvement in comparative accuracy, comprehensiveness and clarity.

WP2: Applying real-time AI workflow analysis for assisting training.
The goal of this stage is to present workflow analysis (surgical steps and instruments) to surgical teams in real-time during an operation and post-operatively to promote reflective practice, and to assess its educational and operational benefits. Our existing ethics approval (IRAS 127474) will facilitate this work. We have already performed pre-clinical (IDEAL Stage 0) studies of the candidate AI model, which is now capable of accurate and automatic surgical phase (F1 score 90%) and step (F1 score 75%) recognition (see Underpinning AI model section)[20, 21, 23].

The next stage of this work is a first-in-human pilot study (IDEAL Stage 1) of the feasibility, acceptability and safety of real-time intra-operative workflow analysis[28, 33]. This would be single centre, recruiting six prospective cases[31]. We will deploy our AI model via the NVIDIA Clara AGX™ system which will receive live surgical video from the surgical endoscope, and display on a second screen the current and upcoming surgical steps and associated instruments. This will be displayed to the surgical assistants, scrub nurses and the wider surgical team but not to the primary operating surgeon. The operative video will be available to the surgeons immediately post-operatively (via the Touch Surgery™ Enterprise online platform), automatically indexed for surgical workflow, with performance metrics available for review (e.g. step frequency, order and duration). Performing surgeons will be presented with this annotated video and associated performance metrics, and guided by expert surgeons to analyse the operative strengths and areas to improve for each case – akin to performance coaching in other high-stakes disciplines[35]. The outcomes of this study will be assessing feasibility (functioning display of intra-op workflow and availability of video metrics post-op), acceptability and perceived educational yield of intra and post-operative workflow display[28, 31, 35]. Safety concerns (e.g. distraction of surgical teams by the AI display) will be actively monitored for and assessed via questionnaires[28]. We will also conduct a human factors analysis to improve how the AI model outcome is displayed via observation and interviews[28]. Data will be collected on a case-by-case basis, guided by the DECIDE-AI guidelines, and used to iteratively improve the AI-human interface[28]. The success criteria for progression to the next stage are based on the primary outcomes - if feasibility (>80% of cases with successful operative metric display), acceptability (>80% of participants finding the intervention acceptable) and educational impact (>80% participants finding the intervention educational) are demonstrated. If the study fails on its first trial, the intervention will be iterated and re-trialled if deemed necessary.

We will then perform a single centre prospective development study (IDEAL Stage 2a) to assess the potential effectiveness and safety of this AI workflow display in 30 prospective cases[28, 31]. The primary outcomes of this study will be assessing the perceived educational value of seeing current and upcoming surgical steps and instruments for all members of the surgical team, and the perceived educational value of the post-operative video-assisted coaching[35]. Both will be assessed via a structured questionnaire[35]. We will also assess for changes in operative time, performance and outcomes when compared to a similar retrospective cohort. Outcomes will be recorded via an established prospectively maintained database[36]. These post-operative outcome metrics will be derived from our previous multicentre collaborative pituitary surgery outcomes study[9-12]. Operative performance will be assessed using independent modified OSAT ratings of each surgical video[37]. Human factors analysis and safety monitoring will continue and iteratively improve the technology interface, again guided by the DECIDE-AI guidelines[28, 31]. Outcomes metrics for all stages are detailed further in the Data Collection section. This study will provide the necessary experience and technological refinement for planning multicentre explorative (IDEAL Stage 2b) and comparative (IDEAL Stage 3) studies in the future[28, 31]. We plan to pursue these (via a dedicated project grant) if the intervention demonstrates any educational impact (>80% of participants find it educational), and it demonstrates an impact on operative performance or postoperative outcomes.

WP3: Applying AI workflow analysis to assist surgical outcome prediction & improve clinical effectiveness.
The codification of surgical videos using workflow analysis enables outcome prediction for each surgery, which can be presented to clinicians post-operatively as a decision-support tool[16, 38-40]. Supported by a previous CRUK predoctoral fellowship and an upcoming EPSRC project grant (see Related Funding section), we have developed and are refining pre-clinical AI models capable of outcome risk profiling through operative video workflow analysis.

The first step in clinical translation is a single centre first-in-human pilot study (IDEAL Stage 1) of six prospective cases[28, 31]**.** After input of the operative video data, the predicted outcome profile will be displayed (via NVIDIA

Clara AGX™) to expert surgeons, trainee surgeons and specialist nurses. As the ultimate goal of this technology in the first instance is to support decision-making post-operatively but pre-discharge, only common inpatient outcomes that were readily quantifiable were included (see Data Collection section)[11, 36]. We will explore the perceived feasibility and acceptability of the AI display, as well as qualitative feedback guided by human factors analysis[28, 33]. This will be used to iteratively improve the AI-surgeon display interface[28, 31]. The success criteria for progression to the next stage are based on the primary outcomes - if feasibility (>80% of cases with successful outcome prediction display) and acceptability (>80% of participants finding the intervention acceptable) are demonstrated. If the study fails on its first trial, the intervention will be iterated and re-trialled if deemed necessary.

Next, a single centre prospective development study (IDEAL Stage 2a) will be performed to assess the potential effectiveness and safety of the AI model display in 30 prospective cases when displayed to expert surgeons, trainee surgeons and specialist nurses looking after the patient[28, 31]. AI model outputs will be made available after the operation and for the duration of the patient's stay. At discharge, each participant will be asked, via structured questionnaire, about the technology's perceived effectiveness (whether it assisted with earlier recognition or prophylactic treatment of complications for high-risk cases, or whether it facilitated earlier safe discharge for low-risk cases) and perceived safety (via human factors orientated interviews)[28]. This study will provide the necessary experience and technological refinement for planning multicentre explorative (IDEAL Stage 2b) and comparative (IDEAL Stage 3) studies, which we plan to pursue via a dedicated project grant in the future[28, 31]. We plan to pursue these (via a dedicated project grant) if the intervention demonstrates satisfactory perceived safety (>95% of participants find it safe) and effectiveness, (>80% of participants find it effective).

Underpinning AI models

Underpinning AI models for the above work packages have been developed and continue to be refined by our team at WEISS[20-23]. During this fellowship, it is likely these models will require iterative changes to improve performance for each of the translational work packages. This will be informed by our study findings, and supported by the continuing opportunistic collection of video and clinical data during these studies. As part of these improvements, we will incorporate methods to enhance the explainability of the AI models, as this will aid acceptance and trust for such clinical AI applications. The continuing support for the engineering infrastructure and personnel to implement any necessary changes to models will be via our existing group funding (see Research Environment section below). For reference, I have included details of the development and evaluation of our AI models below:

After informed consent, each surgical video is uploaded to Touch Surgery™ Enterprise and annotated for surgical steps by expert surgeons[15, 20-23]. The resulting annotated video database (now >200 videos) is split into training and testing datasets (80:20 ratio, 5-fold stratified cross-validation). The videos are sampled at 1 frame per second, serving as the visual input for a two-stage artificial neural network (ANN) which aims to predict the step shown in a given frame. In the first stage, a Convolutional Neural Network (CNN) is pre-trained to recognize steps and phases from a short temporal window (1-5 frames). Due to the ambiguity of the different anatomical and instrument landmarks visible in the dynamic field of view during different steps, a single frame or short sequence may not carry sufficient information to aid correct classification[20]. To compensate for this, a second-stage Recurrent Neural Network (RNN) is trained to improve temporal resolution and prediction consistency. Step recognition was primarily evaluated using F1 scores, calculated per class and then summed across classes, weighted to the prevalence of the respective class. The pilot model achieved a 75% F1 score for step recognition with 50 videos[20], which is continually improving with increasing data input and model refinement.

This model is used in WP2 to recognise intra-operative workflow in real-time and generate performance analytics, for example, surgical step order, duration and frequency. For the operative notes generation (WP1), an additional post-processing stage is implemented to ensure the relationships between steps are clinically consistent, and allow for integration with an operative note template[22]. For the outcome prediction model (WP2), a feed-forward ANN takes the output of the existing CNN-RNN model as an input, in addition to post-operative data, to predict the outcome probabilities for each patient. All ANNs will be trained using an appropriate backpropagation method (e.g. adaptive gradient descent) to minimise the appropriate loss function (e.g. cross entropy). Hyperparameters (including learning rate, and activation function) will be optimised[41]. Final model performance will be compared against logistic regression, with averaged receiver operating characteristic curves created using 10-fold stratified cross-validation. A probability cut-off point of 0.5 will classify the predicted outcome as low or high risk[41].

5.1 Patient recruitment and ethical considerations

All patients and participants will be recruited from a single site National Hospital for Neurology and Neurosurgery (NHNN). For non-patient participants (i.e. the surgical team), all members will be invited to participate if their workflow is affected by the AI intervention in question. Patients will be adults, with pituitary adenomas, undergoing endoscopic transsphenoidal surgery. Only those undergoing revision surgery will be excluded (as this confers very different appearances to the surgical video, which our models are not yet sensitive to). Regarding data diversity, NHNN is the UK's largest pituitary centre, with an international patient base. All consent forms and processes will be available in non-English languages as necessary, and adaptations will be made to maximise inclusiveness if

possible (e.g. audio narration for those who are visually impaired). I have also engaged in numerous public and patient engagement initiatives (see PPI section) to maximise the accessibility of our work.

For the work leading up to this fellowship, we have attained local ethical approval (REC 21/SW/0027, UCL/UCLH as sponsors) and local data governance and IT approvals at the NHNN. These approvals are currently focussed on WP2, and detail the collection of operative videos and associated clinical data, development of AI models, display of workflow analytics to the surgical team, and collection of surgical team feedback (including acceptability, feasibility and usability). Any workflow analytics displayed intra-operatively would not be displayed to the operating surgeon, therefore the risk of distraction is minimised. Thus, WP2 does not need any further amendments and we are satisfied any potential ethical issues have been addressed.

However, WP1 and WP3 would require minor amendments to our existing ethics. For WP1, this would include the display of AI-generated operative notes to assist surgeons in operative note completion. The risk here is low, as there is a human in the loop (the surgeon) who will review any AI output, and will be responsible for any amendments to the note to reflect clinical events. This approximates existing practice at NHNN, where we often use templated operative notes within our electronic medical record system that surgeons amend as appropriate, and therefore we anticipate no/minimal additional risk with AI integration. Similarly, for WP3, we will display a post-operative complication risk profile to clinicians, but again there is a human in the loop, and any management decisions will ultimately be made by the responsible senior decision maker for the patient. Therefore, we anticipate no/minimal additional risk with AI integration.

5.2 Data collection

- Feasibility and acceptability will be assessed via Acceptability of Intervention Measure and Feasibility of Intervention Measure questionnaires (WP1/2/3)[33].
- Human factors analysis will include independent observation and qualitative interview, with any safety concerns analysed using the Systems Engineering Initiative for Patient Safety model (WP1/2/3)[42].
- Educational yield questionnaires will be derived from the Intercollegiate Surgical Curriculum Programme workplace-based assessment proforma (WP2).
- Perceived clinical effectiveness and safety questionnaires (WP3) will be developed specifically for this study.
- Operative performance based on video review will be assessed via a modified Objective structured assessment of technical skill (OSAT) scoring (WP3)[37].
- Inpatient outcomes (WP3) will be collected via our internal prospectively developed database, which aligns with a previous multicentre collaborative study on patient outcomes, and includes the length of stay, hypernatraemia (including diabetes insipidus), hyponatraemia (including syndrome of inappropriate anti-diuretic hormone), new hormone deficit, cerebrospinal fluid leak and visual loss[11, 36]. In the future, we will add longer-term outcomes (such as recurrence, biochemical remission and quality of life) as our dataset becomes more longitudinal and multimodal (e.g. includes radiomics), but this is beyond the scope of the current proposal.

All quantitative data will be descriptively presented, and qualitative data will be thematically analysed. Definite comparative statistics will not be performed during these IDEAL Stage 1 and 2a studies.

Furthermore, during every study, the video and clinical data of each patient will be opportunistically collected to continually improve our underpinning AI models. This will be done after informed written consent, using the secured data collection pipeline detailed below.

5.3 Data management

During the foundational work upon which this fellowship application has been built, I have created a secure data pipeline for the recording and storage of the data. This is instrumental in safeguarding new & existing data used to refine AI models, and new data collected for AI intervention assessment.

Patient demographic and outcomes data are stored via an existing database in XLS format on the NHS secure network drive at NHNN. Patient demographics and outcomes are collected by a senior clinical fellow as part of standard care, following guidelines from a multicentre pituitary surgery outcomes study led by our group to ensure completeness and accuracy[11]. Each record will be given a codified number, to allow linking to the related video, without sharing of any identifiable information. This data will be backed up on an encrypted solid state drive (SSD) which will be kept locked on hospital premises, in the lead supervisor's clinical office.

Surgical videos are recorded from an HD Storz Karl Endoscope across, and are collected using the secure Touch Surgery™ Enterprise platform by Medtronic. The platform fully anonymizes the videos before storing (redacting any footage which is extranasal) and making them available on the secure cloud-based server (region-specific, GDPR compliant, Amazon Web Server). No identifiable patient or user data will be stored on Medtronic servers. Each video will be given a codified number, to allow linking to the related demographics and outcomes data. As

per our Medtronic research contract, we retain full and exclusive ownership rights to uploaded video data as the research team. This data will be backed up on the same encrypted SSD which will be kept locked on hospital premises, in the lead supervisor's clinical office. Anonymised video annotation exports (e.g. workflow, OSATS, etc.) will be stored on an encrypted SSD, and backed up via UCL Data Safe Haven, which offers a secure solution for data storage, handling and analysis. For the intra-operative display of workflow analytics via the NVIDIA Clara AGX™, our AI models are housed on the platform and they will receive a live endoscopic video feed, but no video or clinical data is stored on the platform in the process. AI intervention feedback data (i.e. feasibility, acceptability, human factors, educational yield, perceived safety) will be codified as above and also stored on an encrypted SSD, and backed up via UCL Data Safe Haven. Electronic data will be stored for a minimum of 10 years. Any other identifiable data such as consent forms will be stored in a locked filing cabinet at NHNN for 10 years after study completion.

Concerning data sharing, fully anonymized data will be made publicly available on the WEISS Open Data Server along with the respective publications, AI models, and the policy for data sharing. The PI and supervisors will ask potential data users to provide a brief research proposal detailing how they wish to use the data and the types of data requested. This may form part of a data-sharing agreement. Independent oversight will be sought from the WEISS executive committee. Study participants will be made aware of the data-sharing policies during consent.

5.4 Research environment

This proposal is enabled by the unique interdisciplinary environment at the WEISS UCL, UCL Neuroscience, and NHNN. NHNN is the UK's largest pituitary centre (~150 pituitary adenoma surgeries per year) and UCL Neuroscience is currently rated first in Europe and second in the world by ISI Essential Science Indicators. This neuroscience expertise dovetails with the established AI and clinical innovation expertise at WEISS. WEISS offers a robust infrastructure for clinical translation, including state-of-the-art facilities equipped with cutting-edge resources (e.g. high-performance computing clusters). It also has a dedicated team supporting research ethics, clinical trials, quality management and public engagement. Engineering or technical demands of this proposal (i.e. iterating AI models between IDEAL stages) will be supported by an existing EPSRC AIDPitSurg grant (EP/Y01958X/1, 540000gbp, over 18 months from November 2022). My proposed PhD supervisors are co-investigators and the senior leads on this project, which will fund two computer science postdoctoral research fellows to drive AI model improvement. I am not listed as a co-investigator on this grant, and I am not receiving personal support from this funding. Altogether, the integration of these centres of excellence, with key industry collaborators, provides an exciting opportunity for this proposal to flourish.

5.5 Project management

By the start of this proposed fellowship I would have been working at WEISS, UCL and NHNN for four years – one as a clinical research fellow, followed by three years as an NIHR academic clinical fellow. During this time, I have dedicated my research time to the necessary foundational work needed for this PhD proposal. Furthermore, the last six months of my ACF will be research time spent continuing my ACF projects, which are precursors to my proposed fellowship (see Research Timetable Figure). Throughout these four years, I have been mentored by my proposed PhD supervisors, Mr Hani Marcus and Prof Stoyanov, who are experts in the field, and have established relationships with key collaborators (e.g. IDEAL collaboration) and industry partners (Medtronic and NVIDIA). Their regular supervision (see Research Support Section) and these established partnerships will be critical to delivering the proposed research timetable.

5.6 Output and next steps

Research output from this fellowship will be published in peer-reviewed journals and presented at national and international conferences, leveraging my track record of scientific writing (see Research Background) and research dissemination (see PPI). We will generate datasets of scientific value, which we will make accessible where possible (see Data Management).

If a workstream demonstrates feasibility, acceptability, and potential for clinical impact, we will apply for further funding to carry out IDEAL Stage 2b (multicentre) and Stage 3 (comparative trial) studies to explore the impact on real-world patient outcomes across centres and generate higher levels of evidence. For this, we will seek further project grants and formal Clinical Trials Unit support.

7. Patient & Public Involvement

Please describe how patients/service users, carers and the public have been involved in developing this proposal

I recognise the importance of public and patient involvement (PPI) in research, particularly for topical research such as AI, where I have studied wider perspectives on the technology. Applying this knowledge, I became faculty for the UCL SURGE public engagement project, where I led the creation of an animated video on video analytics AI research, to make it more widely accessible and relatable. More recently, I have received funding from the National Brain Appeal (Small Acorns Fund, £2700) and WEISS UCL (Public Engagement Fund, £2700) to co-lead international consensus studies on patient priorities and core outcomes in pituitary adenoma research. Through this project, we have engaged with hundreds of patients through the Pituitary Foundation using accessible videos (that I co-created), so that on completion, this work will form part of a research translation toolkit focussed on high-priority patient needs. This work will be completed by the time of my proposed fellowship, and the research priority list and core outcome set, encompassing patient views, will be encompassed in the output from the proposed and downstream projects.

For this specific proposal, I engaged with the WEISS UCL Pituitary Patient Advisory Group, which is a growing group of pituitary adenoma patients interested in research, funded via WEISS. Two patient representatives from this group, one of whom is also a representative of the Pituitary Foundation, have reviewed this application and I have integrated their insights whilst refining lay summaries, study design, and project outcomes. In return for their time and effort, they have received fair monetary numeration from WEISS. This application has also been reviewed by the WEISS Public Engagement Manager, Simon Watt, to ensure the PPI is appropriately integrated into this research design, technological iteration and future studies.

Please describe the ways in which patients/service users, carers and the public will be actively involved in the proposed research, including any training and support provided

Throughout my fellowship, I intend to engage with the WEISS UCL Pituitary Patient Advisory Group for the six proposed studies by asking for their insights on each study protocol. This will ensure the study design is sensitive to patient priorities, and that the outcomes are aligned with patient interests. Even though concepts for IDEAL stage 1 and 2a studies will be similar, I believe there is value in repeated active PPI at each stage, given how dynamic the public outlook and views on AI are. For each protocol, at least two members of the group will be approached, and their time will be renumerated appropriately. Similarly, during the amendment of our existing ethics, I will consult the group to ensure any changes to the patient-facing documents are appropriate and understandable.

Furthermore, PPI will be a cornerstone of the dissemination of work produced during this fellowship. I will apply my skills in creating engaging informational videos to generate video summaries of each of the three workstreams, working with the WEISS UCL Pituitary Patient Advisory Group to do so. I will share these research summaries with the Pituitary Foundation so that they are available for interested patient members. Furthermore, I plan to engage with the annual WEISS Science of Surgery event, which aims to disseminate the centre's research, to local and underserved communities in London. I hope to use this as a way to display our AI on a simulated model, with a focus on exploring views and debunking myths about AI in surgery. Similarly, I will build on WEISS's existing collaboration with an interactive theatre company and deliver a scenario-based education tool for surgical video collection and AI, aimed at patients and the public. Finally, I have recently shared our research via media (BBC, Reuters, NowThis and RadioFrance) and could use this experience if the opportunity were to arise in the future.

I am confident that the integration of PPI will not only strengthen this proposal's design, execution and dissemination, but will also help me develop this important skill set for my future work as a surgeon and as a scientist.

If it is considered not appropriate and meaningful to actively involve patients/service users, carers and the public in your proposed research, please justify why

Not applicable, see above.

8. Training & Development and Research Support

Proposed training and development programme

I aim to be a high-quality academic neurosurgeon, with an emphasis on advancing surgical technology and innovation. The natural next step in my development as a clinician-scientist is completing my PhD, to further develop my general academic skills and technical skills (i.e. data science), whilst maintaining my clinical skills. This proposal epitomizes my interests and ambitions, and having the opportunity to pursue it will give me the time, practical experience and formal training to develop these further.

1) Technology Translational Skills:
The majority of my research thus far has focussed on pre-clinical technology development of new technologies (robotics & AI) or clinical studies of established treatments and technologies. During this PhD, a key learning outcome for me is learning how to bridge this gap systematically and safely through early clinical translation. I hope to acquire the necessary technology translation skills in practice and via the learning opportunities available at UCL. For example, the ACCELERATE Potential and ACCELERATE Expertise courses (via the Academic Careers Office) to improve my translational research skills. Working closely with key industry collaborators is a core tenet of technological translation, and I will also build these skills by collaborating with two of the world's largest medical AI companies (Medtronic and NVIDIA). I plan to spend 4-6 weeks on a visiting placement at Digital Surgery at Medtronic (London office), to gain a deeper insight into the processes and perspectives of an industry partner working with clinical academics, to allow me to collaborate more effectively with such partners in the future. Specifically, this would involve joining the research management and product development teams with whom we currently work. Similarly, there are also several entrepreneurship fellowships available in London (e.g. BiteLabs HealthTech fellowship, Newton Fellowship), one of which I will apply for to broaden my translational skill set further.

Furthermore, I plan on visiting an external centre in the USA for a short period (e.g. 4-6 weeks), to learn specific skills in multicentre collaborative AI. I have identified potential hosts at George Washington University, whom we collaborate with, and share interests in neurosurgical AI. Namely, Kevin Cleary, Professor of Bioengineering and Daniel Donoho, neurosurgeon and founder of the Surgical Data Science Collective and the Digital Neurosurgery Conference. They have worked together to perform multicentre AI work in the USA, and as multicentre work is the natural next step of my proposal's AI pipeline after this 3-year fellowship, I believe learning from their expertise would be invaluable for my ongoing development.

2) Data Science & AI skills
Similarly, this proposal will allow me to further my technical skills in data science and clinical AI. Again, I plan to do this in practice, and via courses offered by my institution. For example, the Clinician Coders Workshop (via the Academic Careers Office), to improve my technical data science skills. I also plan to apply to the annual Surgical Data Science Summer School in Strasbourg, with the support of my supervisors, both of whom have been faculty on this renowned data science programme. Furthermore, AI is a rapidly evolving space with near-constant advances in model development. Whilst I have basic coding skills, acquired through experiences such as R statistical software courses via the University of Cambridge, I cannot independently build AI models, and therefore I collaborate closely with AI engineers within our research group at WEISS. I think developing and maintaining this ability in light of the rapidly advancing nature of the field is difficult to achieve whilst balancing clinical, surgical and academic training. Therefore, my approach thus far has been to ensure I have enough practical domain knowledge to communicate effectively within our interdisciplinary team and critically appraise our work. I have done this through online learning platforms (FAST.AI) with the support of my AI engineering colleagues, and I hope to continue to develop these skills through intermittent online and in-person courses during this fellowship.

3) General Academic Skills
Furthermore, through the UCL Doctoral Skills Training Programme, I plan to bolster my wider academic skills in research management, data governance and scientific engagement. I plan to enrol in the UCL Healthcare Engineering Impact Fellowship if time allows, to bolster my skills in leadership, scientific communication, and patient and public engagement. Additionally, I am a faculty member of the UCL AI in Neurosurgery course, and will continue to deliver this course to maintain and develop my teaching skills.

4) Clinical Skills
By the start of this proposed fellowship, I would have finished my ST3 year in neurosurgery. During my PhD, I intend on spending 2 clinical sessions per week (20%) to maintain my surgical and wider clinical skills. This would be in the form of on-call shifts and theatre lists at the National Hospital for Neurology and Neurosurgery. For on-calls, I would take these on an ad-hoc basis, to allow flexibility around my academic work. For theatre lists, I will aim to join elective pituitary surgery lists weekly – for my training and to facilitate the implementation of the proposed studies.

Finally, after my doctoral studies, I plan to apply for an academic clinical lectureship and then academic consultancy upon completion of my training. I intend to subspecialise in pituitary surgery - to align my clinical and academic practice, and allow efficient and bidirectional bench-bedside innovation. This synergy will allow me to keep a patient outcomes centred approach to innovation, and drive further translation of the proposed AI technology to meet this goal in the future. This fellowship therefore optimally integrates my surgical and research interests, and will provide the ideal training for me to grow as a cutting-edge surgeon-scientist.

RESEARCH SUPPORT / SUPERVISOR 1

Name	Hani J Marcus

Institution	University College London

Position	Consultant Neurosurgeon (NHNN) and Honorary Associate Professor (UCL)

Description of support provided
I have established academic and clinical mentors, whose mentorship I have benefitted from over the last few years. I hope to continue to benefit from this mentorship during this proposed doctoral fellowship. Academically, I have been supervised by Mr Hani Marcus since 2016, when I was a visiting third-year medical student from Dublin and he supervised me during a research elective at the Hamlyn Centre, Imperial College London. We executed and published a preclinical validation study of a robotic platform for use in endoscopic neurosurgery. This was an incredibly formative placement for me - igniting my passion for surgical technology and neurosurgery. Following this, Hani has mentored me over the last 8 years, and has been my formal academic supervisor for 4 years at UCL, as a Consultant neurosurgeon at NHNN, an Honorary Associate Professor at UCL, Deputy Director at WEISS, and Lead for Neurosurgery and Brain Cancer for the North Thames NIHR CRN. We have published 41 peer-reviewed papers together, which is a testament to our working relationship, and his supervision skills. His supervision track record includes guiding two students to PhD/MD completion on time, and he is currently supervising 4 PhD/MD candidates. We have achieved this through monthly academic supervision meetings, monthly small-group AI research meetings, and weekly large-group research meetings. I plan to continue this tried and tested schedule during my proposed PhD.

RESEARCH SUPPORT / SUPERVISOR 2

Name	Danail Stoyanov

Institution	University College London

Position	Professor of Robot Vision

Description of support provided
Similarly, I have been supervised by Prof Danail Stoyanov for the last 4 years. He is Professor of Robot Vision, Director of WEISS, Royal Academy of Engineering Chair in Emerging Technologies and Chief Scientist at Touch Surgery (Medtronic). He has been instrumental in my development as a surgical innovator, having published 15 peer-reviewed papers under his tutelage. He is an experienced supervisor, having been involved in the supervision of >45 PhD students, with 19 successfully defended and excelling in relevant fields within academia and industry. He has also been instrumental in forging industry collaborations for this project. If successful, my fellowship would be supported through monthly academic supervision meetings (joint with Mr Hani Marcus, with whom he already co-supervises many PhD students), and by providing the necessary interdisciplinary expertise to execute its goals.

RESEARCH SUPPORT / SUPERVISOR 3

Name	Laurence Watkins

Institution	National Hospital for Neurology and Neurosurgery

Position	Senior Consultant Neurosurgeon (NHNN) and Honorary Senior Research Fellow (UCL)

Description of support provided
Furthermore, I have been supervised by Mr Laurence Watkins since 2021, when I entered run-through neurosurgical training. He is the London Neurosurgery Training Programme Director and a Senior Consultant Neurosurgeon at NHNN. He was my clinical supervisor for my first neurosurgical rotation during training, and has guided me since during my surgical training. We have had scheduled clinical supervision and training meetings at least bi-annually, and he has always been available ad-hoc for training and pastoral support during this time. Resultantly, I have been able to excel in my clinical training thus far, as demonstrated by my multi-consultant feedback reports, multidisciplinary team feedback assessments, surgical logbook numbers and ARCP reports. During this proposed fellowship, I would plan to meet with him at least yearly (but aiming for every 6 months), to ensure support for the maintenance of my clinical skills and support my eventual return to clinical training after my PhD. I plan to use the Intercollegiate Surgical Curriculum Programme Objective-Setting Supervision Meeting template (as used during my previous clinical rotations), to ensure supervision is structured and targeted to all domains of my clinical development.

Collaborations
My proposed fellowship involves two key industry collaborators, with neither company expecting to access the outputs of the research prior to publication. These collaborations are a huge asset to the proposed pipeline (keeping costs relatively low, as they provide hardware and software at no monetary cost), and the future scalability (i.e. to multicentre trials) and eventual translation of this work. The project is primarily supported by a global leader in medical technology, services and solutions, Medtronic. Medtronic (via Touch Surgery™ Enterprise) provide a world-leading secure data management and annotation platform, allowing us to continually develop our state-of-the-art AI models. Touch Surgery™ Enterprise is already in use within the operating theatres at NHNN, with the necessary governance and biomedical approvals in place. They have provided a letter of support as a supplier for this application. Similarly, we will also work closely with the world leader in AI Computing, NVIDIA, who have supplied us with a Clara AGX™ platform and their technical support as part of a related EPSRC grant (see Research Plan). This technology allows state-of-the-art efficient embedded computing systems with display monitors, allowing us to store AI models and present workflow analysis outputs with clinicians. The necessary governance and biomedical approvals are in process for the NVIDIA Clara AGX™ system. Finally, I have also attained a letter of support from Prof Peter McCulloch, Chair of the IDEAL Collaboration and Professor of Surgical Science and Practice at Oxford University, as an academic collaborator whom I intend to seek regular review of my work this fellowship. This will ensure that it adheres to the IDEAL guidelines for systematic and comprehensive surgical technology evaluation.

Host Organisation Support Statement
The following statement of support must be completed by the head of your lead academic department. In order to contribute to this application, this individual must be invited to do so via the 'Participants and Signatories' section of the form.

Statement of support

Dear Committee Members

RE: Danyal Khan – NIHR Doctoral Fellowship

It is with great pleasure that I write this letter of support for Danyal Khan's application for an NIHR Doctoral Fellowship. I write this as Professor of Neuroscience and Head of the Department of Neuromuscular Diseases at UCL Queen Square Institute of Neurology, which Danyal joined three years ago as an NIHR Academic Clinical Fellow.

Balancing surgical and academic training is a challenging feat, however, Danyal has demonstrated excellence across domains as evidenced by his clinical and research track record. Namely, he was the highest-ranking candidate across the UK for neurosurgical national selection, has led a large collaborative pituitary surgery outcomes project, has published 70 peer-reviewed publications, and has attracted numerous grants to generate foundational work for this application.

His work is at the cutting edge of surgical data science, using artificial intelligence (AI) to analyse operative videos and assist with audit, training and prognostication – with pituitary surgery as an ideal exemplar. He has dedicated himself to the field of surgical AI for the last 4 years, integrating seamlessly with an interdisciplinary team at the Wellcome EPSRC Centre for Interventional Surgical Sciences. Not only has he acquired specific skills in clinical AI and data science, but he has also built strong relationships with his clinical, academic and engineering mentors.

UCL, and the Institute of Neurology in particular, is an ideal host for Danyal to continue his academic career. UCL Neuroscience is currently rated second in the world by ISI Essential Science Indicators. In the calendar year 2021, Institute staff published 2197 papers; 85 were published in the top 50 of all scientific journals (ranked by ISI impact factors), including Nature, Science, Lancet, BMJ and JAMA. RAND report shows that UCL has the highest share of highly cited publications in Neurology in England. A large proportion of the Institute's funding is obtained from the Higher Education Funding Council for England. IoN was a major part of REF2021 Uo4 which showed that UCL is number one in the UK for research power in psychology, psychiatry and neuroscience. The Institute currently holds 689 active research projects, totalling £315 million, for research from the principal medical charities concerned with neurological diseases, and from government agencies such as the Medical Research Council, and we also receive significant philanthropic support. There are 143 Principal Investigators at IoN, including: 106 professors/professorial research associates and 26 emeritus professors; 12 Fellows of the Royal Society; 32 Fellows of the Academy of Medical Sciences; 1 Nobel Prize winner.

The Institute and its partner hospital, The National Hospital for Neurology and Neurosurgery, are not only focused on providing the best possible care for our current patients, but also towards developing new therapeutic approaches through which we will provide coming generations of patients with novel, effective treatments. To do this, we have renewed focus on technology development to make our treatments safer and more effective, dovetailing with the NIHR AI research strategies.

Our Department is the ideal place to host this project and to enable Danyal to develop. To support him, we will provide support through a number of mechanisms unique to UCL. During his PhD Danyal will be enrolled into UCL's Doctoral Skills programme to structure his academic and professional development, whilst having the opportunity to engage in complementary courses and services. For example, our Clinical Academic office offers dedicated training in data science, coding and translational research (e.g. ACCELERATE programme). This is accompanied by multiple educational programmes in UCL and NHNN, such as the Future of Neurosurgery seminars, London Brain School, and UCL-Toronto Brain cancer seminars. There are also regular opportunities to showcase research and network with international visitors, such as the Clinical Academic Training Symposium, IoN Annual Symposium, and Victor Horsely lectures. Furthermore, the IoN offers a dedicated mentoring programme to support early career researchers, which will be available to support Danyal in addition to his existing mentors and supervisors.

We look forward to continuing to foster his potential to be an impactful clinical academic. The potential of this work for neurosurgical research is huge and I strongly support Danyal's application.

Yours faithfully

Professor Linda Greensmith

Sobell Head of Department

Department of Neuromuscular Diseases

9. Detailed Budget

The finance section should provide a breakdown of costs associated with undertaking the research as described in the proposal.

1

Staff type	Lead Applicant

Name	Danyal Khan

Employing organisation	University College London Hospitals NHS Foundation Trust

Grade	Neurosurgery ST3

Starting point on scale	55329

Increment Date	

Justification of Costs
Basic clinical salary for London Neurosurgery Specialty Trainee at ST3 level

Details of Posts and Salaries Costs

	Total (£)
Salary (£)	
Geographical Weighting (£)	
Other allowances (£)	
Superannuation (£)	
Nat. Insurance (£)	
Current Annual Costs (£)	**0**

Staff Type: (Lead Applicant)-Mr Danyal Khan
Type of cost: HEI
% Full-time on this research: 100.0
Total months on this research: 36.0
Justification of Cost: My basic salary for ST3 was 55329 in 2023, to which an Apprenticeship Levy has been added. All costs calculated with UCL Research Support Services.

	Year 1	**Year 2**	**Year 3**	**Total Cost (£)**
Basic Salary	55628	55628	55628	166884
National Insurance	7001	7001	7001	21003

Superannuation	8675	8675	8675	26025
Geographical weighting / Other allowances (if applicable)	4500	4500	4500	13500

Totals	Applicant	Support Staff	Shared Staff	Total Cost (£)
Total HEI	181929	0	0	181929
Total NHS	0	0	0	0
Total Commercial	0	0	0	0
Total Other	0	0	0	0

Travel, Subsistence and Conference Fees

Type of cost: HEI
Description: Conference Fees
Conference: True
Item Description: Research dissemination
Justification of Cost: Participation in conferences is essential for disseminating research findings, receiving feedback, networking with peers and establishing collaborations. Therefore, a budget has been allocated for attending two major conferences relevant to the project per year. One of these would be a clinical conference, for example, the European Association of Neurosurgeons Congress or the World Federation of Neurosurgical Societies Congress. The other conference would be an AI or data science conference, for example, the International Conference on Medical Image Computing and Computer-assisted Intervention (MICCAI) or the Digital Neurosurgery Conference. I have budgeted up to 1000 GBP per year to showcase the findings of the project and learn from international communities with shared interests.

	Year 1	Year 2	Year 3	Total Cost (£)
	1000	1000	1000	3000

	Year 1	Year 2	Year 3	Total Cost (£)
Travel, Subsistence and Conference Fees Total	1000	1000	1000	3000

Equipment

Type of cost: HEI
VAT relief to be applied: False
Description: Personal Computer
Price excluding VAT £: 1000
Expected life time: 36
Justification of Cost: I would require a personal computer with the processing capability to manage large video datasets (editing, pre-processing, transfer) and basic statistical and machine learning analysis. For example, a MacBook Pro or AlienWare laptop, which are generally between 1500-2000 GBP in cost. I will use the max cap of 1000 to contribute towards this.

	Year 1 (exc VAT)	Year 2 (exc VAT)	Year 3 (exc VAT)	Total Cost (£) (exc VAT)
	1000	0	0	1000

Type of cost: HEI
VAT relief to be applied: False
Description: Hard drive for data storage
Price excluding VAT £: 500
Expected life time: 36
Justification of Cost: Our operative videos are 4k quality, and on average are 50 gigabytes in size. Therefore, for

the 138 videos collected during this study (WP1 36 videos, WP2 66 videos, WP3 36 videos), we will require 6,900GB of storage. We will store this data on cloud-based and physical encrypted hard drives. The former is provided by the Touch SurgeryTM Enterprise platform (Medtronic), but I would need funding for the latter – for mobile encrypted solid state drives (SSD) to extract videos from endoscope video recorders and act as backup storage. In our experience with UCL Marketplace, such an SSD, with an associated USB-capable case and protective cover, can be purchased for approximately 500 GBP.

	Year 1 (exc VAT)	Year 2 (exc VAT)	Year 3 (exc VAT)	Total Cost (£) (exc VAT)
	500	0	0	500

	Year 1	Year 2	Year 3	Total Cost (£)
Equipment Total	1500	0	0	1500

Summary

	Travel (excl Conferences)	Travel (Conferences)	Equipment
Total HEI	0	3000	1200
Total NHS	0	0	0
Total Commercial	0	0	0
Total Other	0	0	0

Consumables
No costs have been added

Patient and Public Involvement

Description: Renumeration for Study Protocol Review
Type of cost: HEI
Please describe fully: Renumeration for Study Protocol Review
Justification of Cost: Payment of 50GBP for review of each of study protocol by a member of our Pituitary Patient Advisory Group at WEISS UCL. We anticipate a review will take 2-4 hours of time, and will ask two patients to review each protocol. There will be 6 protocols generated during this fellowship (2 studies for each of the 3 work packages).

	Year 1	Year 2	Year 3	Total Cost (£)
	200	200	200	600

Description: Dissemination Video Costs
Type of cost: HEI
Please describe fully: Dissemination Video Costs
Justification of Cost: I will apply my skills in creating engaging informational videos to generate video summaries of each of the three workstreams, working with the WEISS UCL Pituitary Patient Advisory Group to do so. From experience, each video costs 2500-3000GBP when contracting an animator. I will share these research summaries with the Pituitary Foundation, so that they are available for interested patient members.

	Year 1	Year 2	Year 3	Total Cost (£)
	3000	3000	3000	9000

	Year 1	Year 2	Year 3	Total Cost (£)
Patient and Public Involvement Total	3200	3200	3200	9600

Other Direct Costs

Description: Data annotation costs
Type of cost: HEI
Please describe fully: Data annotation costs
Justification of Cost: Crucial to any AI technology is data curation and management. To create the video AI workflow models employed in our AI interventions, we had to generate videos annotated with steps and instruments. The accuracy of this annotation is crucial as it provides the ground truth for these supervised learning models. However, this is time-consuming and requires technical expertise. We have established relations with 3rd party professional medical data labellers (Anolytics), who have provided consistent, high-quality, high-throughput labelling for our dataset. During this proposal, we aim to translate these AI models and perform early clinical evaluation. In our work packages, we will 1) continually assess the accuracy of the models being delivered in our intervention, 2) use video assessment as a study metric (WP2 only), and 3) opportunistically collect videos which will be annotated and used to improve the underpinning AI models further. For 1), to assess the accuracy of the underpinning AI models, we must provide ground truth labelling for the video cases that the AI workflow model will be analysing. This amounts to 36 videos (6 for IDEAL stage 1, 30 for IDEAL stage 2) per WP, resulting in 108 total videos. Our mean video length is each video is 78.5 minutes (i.e. 4710 seconds) in length, resulting in 4710 frames to be annotated per video, when sampled at 1 frame/second. The total number of frames for annotation is 508,680 (4710 x 108). We have been quoted a price of 0.02 USD per frame from Anolytics, therefore the cost of ground truth annotations here is 10,174 USD. For 2), one of the outcomes of WP2 is to compare the change of performance (using the validated OSAT scoring system) in an AI intervention cohort (n=36; 6 for IDEAL stage 1 and 30 for IDEAL stage 2) and a retrospective control cohort (n=30). This means 66 videos must be annotated for OSAT scores, which we have elected should be done independently by expert annotators (and their scores averaged). We have received a quote of 50 USD per video from Anolytics for this, and have been satisfied with their annotation expertise for this task on previous projects. This results in an additional 3300 USD (66 x 50). For 3), this will be a byproduct of the annotations of 1) and 2), at no additional cost. Therefore, in total 13474 USD will be required for data annotation, which is approximately 11736 GBP (or 3912 GBP per year across 3 years) after a 10% buffer is applied to contemporary exchange rates. By outsourcing these annotations, we can improve the accuracy and standardisation of our data curation. Furthermore, this process is otherwise very time-consuming and has limited educational benefit beyond the first few cases. Therefore outsourcing it will give me a lot more time to spend on training opportunities and executing the actual proposed studies.

	Year 1	Year 2	Year 3	Total Cost (£)
	3912	3912	3912	11736

	Year 1	Year 2	Year 3	Total Cost (£)
Other Direct Costs Total	3912	3912	3912	11736

Training And Development

Type of cost: HEI
Description: Applicant PhD Tuition Fees
Item Description: UCL PhD Tuition Fees
Justification of Cost: UCL PhD Tuition Fees

	Year 1	Year 2	Year 3	Total Cost (£)
	6035	6215	6400	18650

Type of cost: HEI
Description: Overseas Research Visits
Item Description: Overseas Centre Training Fellowship
Justification of Cost: In addition to the local training gained through UCL's existing courses and through learning in practice during this PhD, I would also like to gain an alternative experience via an external overseas centre. I have spent the last four years working within my current research group, and whilst this has been very productive and enjoyable, I believe I would benefit hugely from the opportunity to broaden my perspectives and spend a short period, for example, a 4-6 week research fellowship, in another centre. I have identified potential hosts at George Washington University, whom we collaborate with, and share interests in neurosurgical AI. Namely, Kevin Cleary, Professor of Bioengineering and Daniel Donoho, neurosurgeon and founder of the Surgical Data Science Collective and the Digital Neurosurgery Conference. They have worked together to perform multicentre AI work in the USA, and as multicentre work is the natural next step of my proposal's AI pipeline, I believe learning from their expertise would be particularly valuable to my ongoing development.

	Year 1	Year 2	Year 3	Total Cost (£)
	0	4000	0	4000

Type of cost: HEI
Description: Overseas Research Visits
Item Description: Industry Training Fellowship
Justification of Cost: I plan to spend 4-6 weeks on a visiting placement at Digital Surgery at Medtronic (Ireland or London office), to gain a deeper insight into the processes and perspectives of an industry partner working with clinical academics, to allow me to collaborate more effectively with such partners in the future.

	Year 1	Year 2	Year 3	Total Cost (£)
	0	1000	0	1000

Type of cost: HEI
Description: Short Courses
Item Description: Entrepreneurship Programme
Justification of Cost: There are also several paid entrepreneurship fellowships available in London (e.g. BiteLabs HealthTech fellowship, Newton Fellowship), one of which I will apply for to broaden my translational skill set further.

	Year 1	Year 2	Year 3	Total Cost (£)
	0	0	1000	1000

Type of cost: HEI
Description: Workshops
Item Description: UCL Clinician Coders Workshop
Justification of Cost: I will attend the the Clinician Coders Workshop (via the UCL Academic Careers Office), to improve my technical data science skills. https://www.ucl.ac.uk/health/academic-careers-office/training-portfolios/data-arcade/clinician-coders

	Year 1	Year 2	Year 3	Total Cost (£)
	350	0	0	350

Type of cost: HEI
Description: Short Courses
Item Description: Surgical Data Science Summer School in Strasbourg
Justification of Cost: I also plan to apply to the annual Surgical Data Science Summer School in Strasbourg, with the support of my supervisors, both of whom have been faculty on this renowned data science programme. If successful, I am sure this will further bolster my data science skills, specifically within the realm of computer vision. https://www.edu4sds.org/#Program

	Year 1	Year 2	Year 3	Total Cost (£)
	300	0	0	300

	Year 1	Year 2	Year 3	Total Cost (£)
Training And Development Total	6685	11215	7400	25300

Total Direct Costs

	Year 1	Year 2	Year 3	Total Cost (£)
Total Direct Costs	75218	78548	74733	228499

Indirect costs

HEI Indirect costs

Type of cost: Other Indirect Costs
Name of Higher Education Institute: UCL
Justification of Cost: UCL Indirect Costs provided by research support services team

	Year 1	Year 2	Year 3	Total Cost (£)
	60753	60753	60753	182259

Type of cost: Estates Charges
Name of Higher Education Institute: UCL
Justification of Cost: UCL Estate Costs provided by the research support services team

	Year 1	Year 2	Year 3	Total Cost (£)
	26846	26846	26846	80538

	Year 1	Year 2	Year 3	Total Cost (£)
HEI Indirect costs Total	87599	87599	87599	262797

Commercial Indirect Costs
No costs have been added

Other Partner Organisation Indirect costs
No costs have been added

	Year 1	Year 2	Year 3	Total Cost (£)
Total Indirect Costs	70079	70079	70079	210237

Research Costs Requested From Funder

	Direct costs	Indirect costs/Overheads	Amount requested
Total Higher Education Institution Costs (80%)	200199	210237	410436
Total Higher Education Institution Costs (100%)s	28300	0	28300
Total Commercial Costs	0	0	0
Total Other Parther Organisation Costs	0	0	0
Total NHS Research Costs	0	0	0
Total Research Costs Requested From Funder	228499	210237	438736

Summary

	Year 1	Year 2	Year 3	Total Cost (£)
Salaries	**60643**	**60643**	**60643**	**181929**
Travel-subsistence-and-conference-fees	**1000**	**1000**	**1000**	**3000**
Equipment	**1200**	**0**	**0**	**1200**

Consumables	**0**	**0**	**0**	**0**
Patient-and-Public-Involvement	**2560**	**2560**	**2560**	**7680**
Other-Direct-Costs	**3130**	**3130**	**3130**	**9390**
Training And Development	**6685**	**11215**	**7400**	**25300**
HEI-Indirect-costs	**70079**	**70079**	**70079**	**210237**
Commercial-Indirect-Costs	**0**	**0**	**0**	**0**
Other-Partner-Organisation-Indirect-costs	**0**	**0**	**0**	**0**
Total	**145297**	**148627**	**144812**	**438736**

NHS Support Costs

Cost Item:	Year 1	Year 2	Year 3	Total Cost (£)
NHS Service Support Cost	4802	4802	4802	14406.0

Justification:
NHS Service Support Cost: As per SoECAT, in discussion with the local ACORD specialist.

	Year 1	Year 2	Year 3	Total Cost (£)
NHS Support Costs Total	4802	4802	4802	14406

Have you discussed and agreed these costs with the Lead Network?	Yes

NHS Treatment Costs

Cost Item:	Year 1	Year 2	Year 3	Total Cost (£)
NHS Excess Treatment Cost / Cost saving	0	0	0	0.0

Justification:
NHS Excess Treatment Cost / Cost saving: As per SoECAT, in discussion with the local ACORD specialist.

	Year 1	Year 2	Year 3	Total Cost (£)
NHS Treatment Costs Total	0	0	0	0

Have you discussed and agreed the NHS costs with the Lead Trust?	Yes

Is the patient/service user care being provided different from the usual standard treatment for the condition?
No

Usual Treatment Costs

Cost Item:	Year 1	Year 2	Year 3	Total Cost (£)
Not applicable	0	0	0	0.0

Justification:
Not applicable: The patient care intervention under investigation is in addition to usual care. There are no

deviations to clinical care otherwise in the proposed projects.

	Year 1	Year 2	Year 3	Total Cost (£)
Usual Treatment Costs Total	0	0	0	0

Excess Treatment costs/(savings)

	Cost per patient(£)	Year 1	Year 2	Year 3	Total Cost (£)
Total NHS Excess Treatment costs/(savings)	0	0	0	0	0

	Total (£)
NHS Support Costs requested from networks	14406
NHS Excess Treatment costs/savings	0
TOTAL NHS SUPPORT AND TREATMENT COSTS	14406

Total Funding Required (including non-research costs)

	Total (£)
Total Research Costs Requested from Funder	438736
Total NHS (Support and Excess Treatment) Costs	14406
Total Funding Required	453142

10. Uploads

(References - References.b.pdf) is included as an appendix within this file.

(Figures / Tables - Figure 1.pdf) is included as an appendix within this file.

(Research Timetable - Research Timetable Figure.d.pdf) is included as an appendix within this file.

(CTU Letter of Support - 04 Supporting documents compiled.pdf) is included as an appendix within this file.

(SoECAT Template - Funder Export-127474_SoECAT.xlsx) is included as an appendix within this file.

11. Participants and Signatories

Devolved Administrations

no Devolved Administrations Participant has been added

Doctoral Primary Supervisor

Doctoral Primary Supervisor Participant

Title	Mr
Forename(s)	Hani
Surname	Marcus
Position	Consultant

Supervisor

Supervisor Participant

Title	Professor
Forename(s)	Danail
Surname	Stoyanov
Position	Professor

Host Organisation - Heads of Department

Head of Department or Senior Manager

Title	Ms
Forename(s)	Debbie
Surname	Hadley
Position	

Administrative Authority or Finance Officer

Administrative Authority or FO Participant

Title	Mrs
Forename(s)	Kasia
Surname	Haremza
Position	Assistant Director of Research Development and

12. Acknowledgement, Review and Submit

Conflict checks:
Please declare any conflicts or potential conflicts of interest that you may have in undertaking this research, including any relevant, non-personal & commercial interest that could be perceived as a conflict of interest.

No conflicts of interest to declare.

Agreement to Terms and Conditions:
I confirm that I have secured all necessary licences and ethical approvals in relation to the research and will abide by the terms of those licences and approvals in carrying out the research.

I confirm that I have read the NIHR Carbon Reduction Guidelines and, where possible, taken steps to reduce the carbon emissions generated by this research.

I confirm that all material within this application, with the exception of sections that other participants are required to complete, is my own work and I have taken steps to ensure that no parts of the application have been inappropriately plagiarised.

I undertake to conduct the research according to such guidelines for good research practice as the Host Organisation may from time to time laydown according to the NIHR Research Governance Framework.

If you are applying for an NIHR award which comes with partnership funding from another organisation (e.g. a charity or industry partner) your information (including your NIHR Application Form and subsequent progress/annual reports) may be shared with the partner organisation as part of the review process and ongoing monitoring of the award.

In completing this, I declare that, to the best of my knowledge, the information provided in this application, is true, accurate and complete. I have read the Guidance Notes and agree to accept the process by which an application is assessed and agree to abide by the conditions under which an award may be granted.

☒Confirmed

Appendices

1. References.b.pdf
2. Figure 1.pdf
3. Research Timetable Figure.d.pdf
4. 04 Supporting documents compiled.pdf
5. Funder Export-127474_SoECAT.xlsx

Appendix #1: References.b.pdf

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Appendix #2: Figure 1.pdf

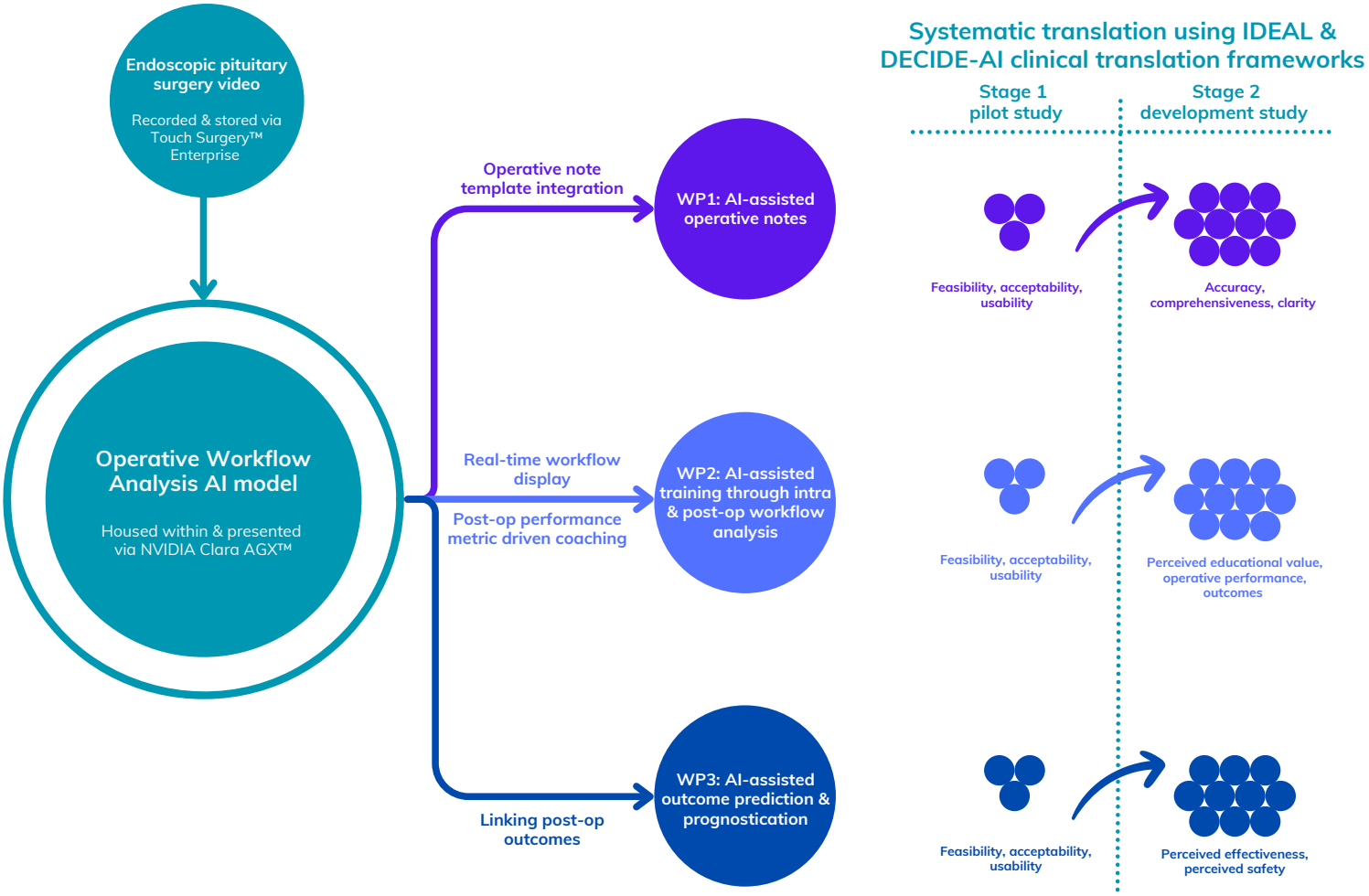
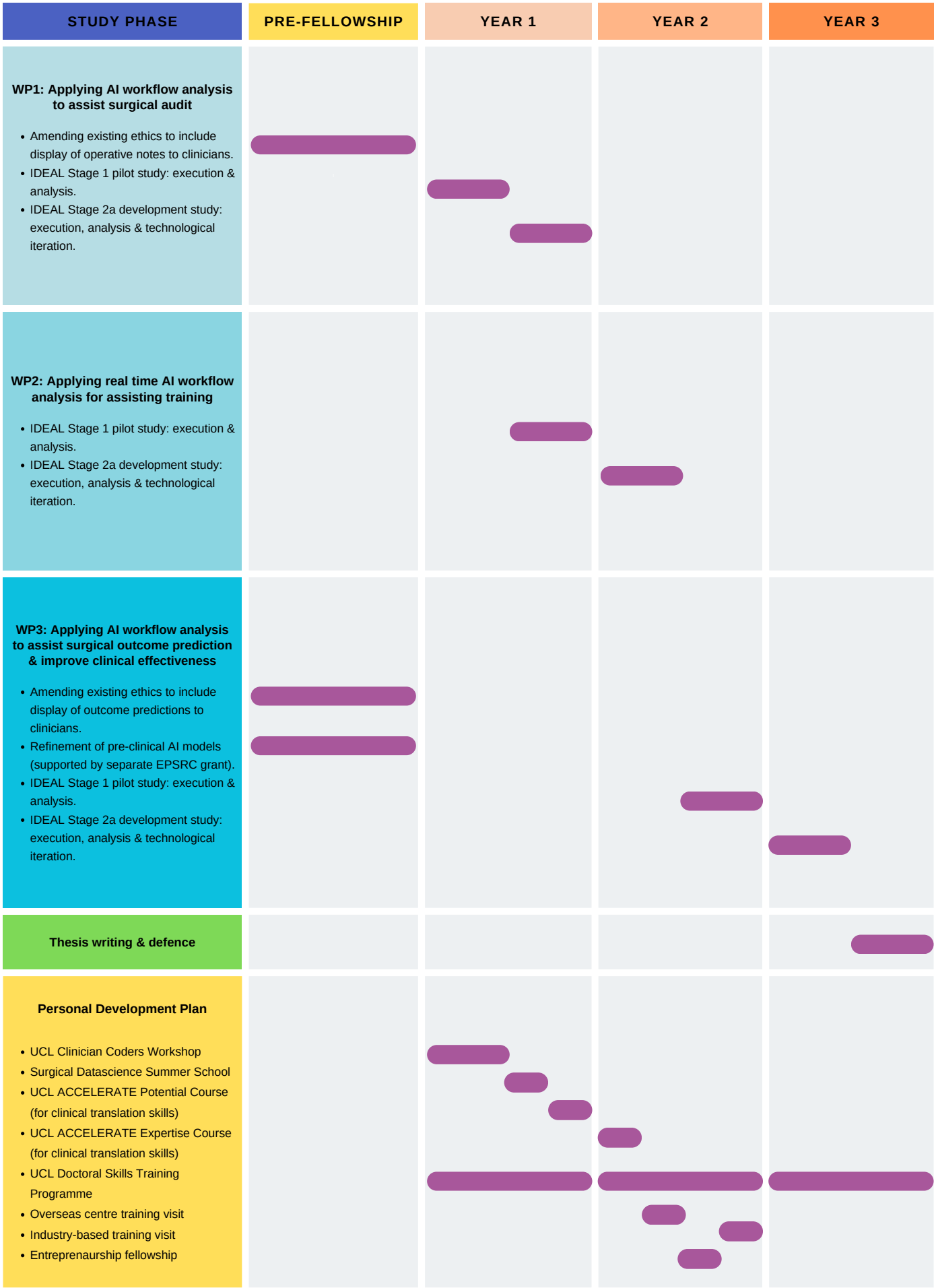


Figure 1: Proposal overview, with three parallel work packages stemming from pre-clinical AI-powered workflow analysis models, and study outcomes per stage.

Appendix #3: Research Timetable Figure.d.pdf



Research Timetable Figure: Gantt chart display of project stages and milestones. The pre-fellowship period represents the last six months of my ACF, which will be protected research time spent continuing my ACF projects, which are precursors to my proposed fellowship. Our centre performs 150 pituitary operations per year (therefore 12-13 per month) so patient recruitment is not an anticipated issue. However, an average of six months per study has been assigned to allow iterative improvements in the AI interface, and give buffers for implementation.

Appendix #4: 04 Supporting documents compiled.pdf

Towards AI-Assisted Pituitary Surgery – The Systematic Translation of Surgical Workflow Analysis to Improve Training, Audit and Prognostication.

NIHR Doctoral Fellowship Application – Round 11

I have sought advice from the Research Support Service Hub delivered by Imperial College London and Partners for crafting this proposal.

We agreed that CTU support was not necessary for the early-stage projects proposed in this application, particularly given the existing infrastructure in place via WEISS UCL and UCLH, and the advisory support offered by the IDEAL colloquium.

Here, I attach a series of supporting documents for this proposal:

- Letter of support from my primary supervisor, Mr Hani Marcus.
- Letter of support from my secondary supervisor, Prof Danail Stoyanov.
- Letter of support from Prof Peter McCulloch, lead of the IDEAL colloquium.
- Letter of support from our primary industrial collaborator, Medtronic.
- Letters of support from Prof Keven Clearly and Dr Daniel Donoho to support an overseas training fellowship.
- Quote for data annotation services from our established 3rd party vendor.
- Local trust approval of the SoECAT (already approved with our local ACORD specialist) and study proposal.

Happy to provide any more information or supportive documents needed.

Yours sincerely,

Danyal Khan

Neurosurgery Trainee (ST3) | London Deanery

Academic Clinical Fellow | University College London

UCL Queen Square Institute of Neurology

The National Hospital for Neurology and Neurosurgery
Queen Square
LONDON
WC1N 3BG

Division of Neurosurgery

NIHR Doctoral Fellowship Committee
21st January 2024

To Whom It May Concern,

I am writing to offer my strongest support for Danyal Khan's application for the NIHR Doctoral Fellowship as a Consultant Neurosurgeon at the National Hospital for Neurology and Neurosurgery, an Honorary Associate Professor at UCL Queen Square Institute of Neurology, Deputy Director at the Wellcome/EPSRC Centre for Interventional and Surgical Sciences (WEISS), and Lead for Neurosurgery and Brain Cancer for the North Thames CRN at the National Institute for Health Research (NIHR).

I've had the pleasure of knowing Danyal for almost a decade, starting when he was a medical student at Imperial College and I was completing my PhD in digital surgery. Our professional relationship has evolved; I've been his academic and clinical supervisor during his Academic Clinical Fellowship (ACF) at NHNN and continue to serve as his academic supervisor.

Danyal is the best trainee I've ever mentored. His research capabilities are outstanding, but what truly sets him apart is his character. He led the largest cohort study of patients undergoing pituitary surgery, coordinating with 30 centres across the UK. This is no small feat; getting that many neurosurgeons to collaborate is a testament to Danyal's interpersonal skills. In addition to being a great team player, Danyal is incredibly hardworking. His dedication to his craft has led him to be ranked first in neurosurgery in the UK. Academically, he is extremely capable, consistently demonstrating a high level of competence and dedication, as evidenced by his very high research output.

Danyal will be based in my research group at WEISS, a multidisciplinary centre with extensive support not only from other clinicians and engineers, but also from experts in regulation, public engagement, human factors, and more. My group consists of a diverse team of 15 researchers and has published 43 papers this year alone, including in high-impact journals such as Nature Endocrinology Reviews and Nature Biomedical Engineering.

Danyal will benefit from robust support and training, including weekly supervision meetings. My track record includes guiding two students to PhD/MD completion on time. One of them, an engineer, went on to attract VC funding for a spin-out company. The other, a clinician, secured a consultant surgeon position and remains in a postdoc role in our group.

In summary, Danyal Khan is an exemplary candidate for this fellowship. His skills, character, and our supportive research environment at WEISS make him an ideal fit. I wholeheartedly endorse his application and eagerly anticipate his continued contributions to surgical research.

Mr Hani J Marcus PhD FRCS
Consultant Neurosurgeon, National Hospital for Neurology and Neurosurgery
Honorary Associate Professor, UCL Queen Square Institute of Neurology
Deputy Director, Wellcome / EPSRC Centre for Interventional and Surgical Sciences
Lead for Neurosurgery and Brain Cancer (North Thames CRN), National Institute for Health Research
Email: h.marcus@ucl.ac.uk

NIHR Doctoral Fellowship Committee
20th January 2024

Dear Committee Members,

I am delighted to offer my strong support for Danyal Khan's application for an NIHR Doctoral Fellowship.

I am a Professor of Robot Vision in the Department of Computer Science at University College London, Director of the Wellcome/EPSRC Centre for Interventional and Surgical Sciences (WEISS), and Royal Academy of Engineering Chair in Emerging Technologies. I am also a Fellow of the Institute of Engineering and Technology, a Fellow of the Royal Academy of Engineering, and a Fellow of the MICCAI Society. I am Chief Scientist at Digital Technologies, Medtronic, and co-founder of Odin Vision Ltd (an Olympus Company), both companies specialising in developing AI products for interventional healthcare. I have over 20 years' experience in developing robotics and artificial intelligence to improve minimally invasive surgery, and have authored and co-authored >500 scientific papers and >25 patents.

I have supervised Danyal since he joined WEISS four years ago, initially as a Clinical Research Fellow, and now as an NIHR Academic Clinical Fellow with an interest and aptitude for neurosurgical applications of artificial intelligence (AI). During his time, he has systematically explored clinical unmet needs in pituitary surgery and has collaboratively developed AI video analysis solutions to address these needs. He has grown into an exceptional young clinical academic, having published extensively in this field already, and integrated seamlessly within our multidisciplinary environment. I am confident he has garnered the necessary skills to deliver this proposal and I am confident that he will become a future research pioneer in this field as I would rate him as one of the brightest talents that I have had the pleasure to work with.

Danyal's proposal focuses on early clinical translation of this AI video analysis technology to improve surgical training, audit, and outcomes. It builds on preliminary work by our group, which I co-supervise with my clinical neurosurgical colleague Mr Hani Marcus, as part of the BRAIN (Brain Surgery with Robotics, Artificial Intelligence and Neuronavigation) and Surgical Robot Vision (SRV) research groups at UCL. This is truly multidisciplinary work which leverages the world-leading AI and clinical neurosurgery expertise at WEISS. At WEISS, engineers, clinicians, and computer scientists work together to develop technologies that enable safer and more effective treatments for patients across a wide range of conditions. In its interdisciplinary synergy, it brings together experts in the fields of surgical and interventional sciences across medical and engineering departments at UCL. The infrastructure at WEISS is unique in the national and international landscape with two dedicated mock operating rooms equipped with endoscopic and laparoscopic towers, multiple ultrasound systems, mobile X-Ray and interventional CT imaging, surgical tracking and navigation platforms, and three full surgical robots dedicated to research use. This infrastructure of course needs brilliant young scientists to maintain and further develop the UK's leading position in computer-assisted interventions, surgical robotics and devices, and AI for healthcare. I believe Danyal is one of these leaders who will be a key contributor to making this happen with a neurosurgical focus.

Every project at WEISS has a strong patient focus. This current proposal dovetails with this goal, offering an exploration of multiple avenues of potential clinical impact. We believe this technology has huge potential benefits, across multiple domains, and is not limited to pituitary surgery alone. Danyal's proposal is a critical steppingstone to unlocking these benefits, placing patient safety and systematic multifaceted assessment at the forefront before future multicentre dissemination of this technology. This fits perfectly with the Centre's translational maxim and also longstanding track record and strategic focus.

If Danyal is successful, we would be delighted to support this project and provide the necessary interdisciplinary expertise and infrastructure to execute its goals. He will be supported through weekly joint supervisor meetings and be afforded abundant training opportunities in data science, clinical translation and research management via WEISS and

wider UCL programmes. My track record includes leading large investments in addition to receiving three research fellowships and the Royal Academy of Engineering Chair in Emerging Technologies. I have been involved in the supervision of>45 PhD students, with 15 successfully defended and excelling in relevant fields within academia and industry.

In summary, I strongly support Danyal's application for this prestigious fellowship. He is a very promising talent who would flourish as a researcher and academic, and I look forward to continuing to supervise and support his development as an effective surgeon-scientist.

Yours sincerely,

Professor Danail Stoyanov, PhD, FREng, FIET, FMICCAI
Professor of Robot Vision, University College London
Director, Wellcome/EPSRC Centre for Interventional & Surgical Sciences (WEISS)
Royal Academy of Engineering Chair in Emerging Technologies

NUFFIELD
DEPARTMENT OF
SURGICAL SCIENCES

UNIVERSITY OF
OXFORD

Peter McCulloch
Professor of Surgical Science & Practice
University of Oxford,
Nuffield Department of Surgical Sciences
John Radcliffe Hospital
Oxford, OX3 9DU
peter.mcculloch@nds.ox.ac.uk

NIHR Fellowships Committee
27th January 2024

Dear Colleagues,

I am happy to recommend Mr Danyal Khan for the NIHR Doctoral Fellowship he has applied for.

I led the DECIDE AI group which developed the evaluation guidelines for early clinical AI which he proposes using, and I think his proposed project is of real significance to the rapidly evolving science of properly evaluating AI in healthcare applications.

I also know Danyal from his contributions to a major piece of work on the evaluation of surgical robots which I co-led with his supervisor over the last two years. This work has recently been published in a major scientific journal, Nature Medicine, and Danyal showed enormous energy and determination in his contribution to getting it written, as well as excellent communication skills and great initiative.

The group he is working in is one of the most dynamic and productive in robotic and high-tech innovation in healthcare in the UK, and he is being mentored and supported by an extremely high-quality faculty with all the resources to make the project a success.

Yours Sincerely,

Peter McCulloch

Medtronic
Touch Surgery™ Enterprise
Digital Technologies
230 City Road
London, UK
EC1V 2QY

11th December 2023

To Whom It May Concern

Application No: R1133154
Project Title: Towards AI-Assisted Pituitary Surgery – The Systematic Translation of Surgical Workflow Analysis to Improve Training, Audit and Prognostication.
PI: Danyal Khan (fellow), Hani Marcus (supervisor) & Danail Stoyanov (supervisor)

I am delighted to write in support of Mr Danyal Khan's proposal for this fellowship. Mr Khan's proposal focuses on adapting AI models, capable of recognising and analysing surgical workflow in endoscopic pituitary surgery, for clinical use in order to assist surgeons in the OR, generate post-operative analytics, and derive performance metrics. This firmly aligns with our vision to provide tools for better intra-operative assistance resulting in quality improvements in the operating theatre. Furthermore, this will build on our existing collaboration on surgical workflow, analytics, and team performance that we have with the Wellcome/EPSRC Centre for Interventional & Surgical Sciences (WEISS) at University College London (UCL). Such industry-university research collaborations are crucial to explore the potential for shaping the best practices in the design, development, and deployment of AI as well as harnessing and training new talent in the field.

Digital Technologies, Medtronic, is one of the world's first companies to use AI to assist with surgical performance, pre-, intra-, and post-operatively. Based on our prior track record in surgical and digital technology through the Touch Surgery™ medical application (over 5 million downloads worldwide), scientifically proven to improve surgical training using cognitive task analysis and gamification elements, we have developed an intelligent intraoperative system for surgery, Touch Surgery™ Enterprise. It enables automated surgical video upload directly from the operating room with real-time safeguarding of sensitive information. Surgical users can organise their videos on our private and secure video management platform powered by AWS. Automatic analytics of the surgical video (for example, surgical phase timings) are then provided to the users. This information can be accessed instantly on both mobile and web applications for shared learning and performance reviews. Furthermore, our solution enables sharing of annotated videos, and viewing through our Touch Surgery™ app and web interfaces, allowing users to have access to best practice and surgical techniques for educational purposes, from across the globe.

Digital Technologies, Medtronic, is delighted to offer a product loan, in-kind, in the form of the following for the duration of the project:

Direct Contribution to Research Project		
	Description	**Value**
Equipment/Materials	Equipment (DS1 computer and associated hardware and software)	£50,000

The supervisory team brings together world-leading experts from a range of disciplines, with each bringing their strong track record of high-quality research programmes and translation. I have full confidence that you will provide vital contributions to the advancement of this field under the leadership of Mr Hani Marcus and Professor Stoyanov with their internationally renowned partners from across the UK.

We look forward to collaborating with Mr Khan to help make the proposed initiative a success.

Yours faithfully,

Karen Kerr, PhD
Core Team Leader – AI & Analysis, Digital Technologies, Medtronic
karen.kerr@medtronic.com

Danyal Khan MB BCh BAO MRCS
University College London
8 January 2024

Dear Dr. Khan:

It is a great pleasure to provide this letter of support for your application to develop workflow analysis for pituitary tumor surgery. As you know from our interactions over the last year, we have been doing similar work in our hospital in collaboration with my neurosurgical colleague Daniel Donoho MD. This work was featured in our recent publication in the International Journal of Computer Aided Radiology and Surgery and I believe this work is very synergistic with the research plan you have proposed.

Therefore I am delighted to provide this open letter of invitation for you to visit our hospital and research group at a mutually convenient time. As the Associate Director of the Sheikh Zayed Institute for Pediatric Surgery Innovation, I am happy to make the resources of the institute available to you, including space, computer resources, and access to observe clinical cases.

I reviewed your research plan and I think it is very innovative, timely, and will advance the state of the art in this important area. As we move towards the digital age of surgery, I believe workflow analysis can serve as a springboard for many improvements which could provide better patient care.

If I can provide any further information, please do not hesitate to contact me. I look forward to welcoming you during your visit.

Sincerely,

Kevin Cleary, PhD
Associate Director for Engineering
The Sheikh Zayed Institute for Pediatric Surgical Innovation
Children's National Hospital
Washington, D.C 20010
Office Phone: 202-476-3809
Cell Phone: 202-309-9582
Email: kcleary@childrensnational.org

Center for Neuroscience
and Behavioral Medicine

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Office: (202) 476-3020
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Emergency: (202) 476-5000

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Division Chief
Professor

John S. Myseros, M.D.
Professor
Division Vice-Chief

Chima Oluigbo,M.D.
Associate Professor

Hasan Syed, MD
Associate Professor

Daniel Donoho, MD
Assistant Professor

Carl J. Bailey, PA-C
Physician Assistant

Robert Crowder, PA-C
Physician Assistant

Stephanie Gurevich, PA-C
Physician Assistant

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Pediatric Nurse Practitioner

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Roger J. Packer, M.D.
Eugen Hwang, M.D.
Lindsay Kilburn, M.D.
Brian Rood, M.D.

Epileptology
William Davis Gaillard, M.D..

Neuroradiology
L. Gilbert Vezina, M.D.

Neurointerventional Radiology
Monica Pearl, M.D.

David C. McCullough, M.D.
(1935-1992)

Danyal Khan MD
University College London

12 September 2023

Dear Dr. Khan,

It is my pleasure to continue to support your work in surgical AI research by extending an invitation to visit our center at Children's National Hospital and the Sheikh Zayed Center for Pediatric Surgical Innovation in Washington, DC USA.

I have been impressed by your prior work and future research plans, as well as your career development process. Your generation of surgeon-scientists are building a unique skill set in translational computer science research to discover applications of artificial intelligence systems in the operating room. Your work is on course to create new datasets, technologies, and techniques for future research as well as direct, patient facing technologies that improve surgical care.

Our center's vibrant work in the area of AI across all aspects of surgical care includes preoperative neuroimaging, intraoperative video assessment, and postoperative outcomes assessment. Your visit could provide both resources, experiences, interactions and new perspectives that may be of value to your current course of study and training. I look forward to welcoming you to Washington, DC.

Respectfully,

Daniel A. Donoho, MD
Pediatric Neurosurgeon
Children's National Hospital

Assistant Professor of Neurosurgery and Pediatrics
School of Medicine and Health Sciences
George Washington University

ddonoho@cnmc.org
Academic Office: (202) 476-1548
Clinical Office: (202) 476-3020

QUOTE

Date: 12th Sept, 2023
Quotation # CT2123
Expiry: 30th Sept, 2023

From:

AnoLytics
A - 83, Sector - 2
Noida – 201301, U.P.
India

To:

Danyal Khan
Department of Neuromuscular Diseases
Institute of Neurology, University College London
London, UK

Description	Qty	Unit Price (USD)	Line Total (USD)
Data Annotation Services (Steps frame-based annotations)	508680	$0.02	$10,173.60
Data Annotation Services (OSAT Project - 66 videos x 1 annotator)	66	$50.00	$3,300.00
Total Amount (USD)			$13,473.60

Terms and Conditions

- 100% advance payment
- Project completion timeline: 4 weeks
- A dedicated project manager will be provided for day-to-day communication.

Payment Details

- Name of the Account Holder: AnoLytics
- Company Address: A-83, Sector-2, Noida-201301, U.P., India
- Name of Bank: Canara Bank
- Account No: 1177201004521
- Routing No: 110015037
- Swift Code: CNRBINBBDOB
- IFSC Code: CNRB0001588

If you have any questions concerning this quotation, pls contact rohan@anolytics.ai

Thank you for your Business!

From: GYEKYE, Michelle (UNIVERSITY COLLEGE LONDON HOSPITALS NHS FOUNDATION TRUST) michelle.gyekye@nhs.net
Subject: RE: EDGE ID 163765- NIHR PhD Fellowship Application - SoECAT query
Date: 24 January 2024 at 11:42
To: SWAIN, Suzanne (UNIVERSITY COLLEGE LONDON HOSPITALS NHS FOUNDATION TRUST) suzanne.swain1@nhs.net, KHAN, Danyal (UNIVERSITY COLLEGE LONDON HOSPITALS NHS FOUNDATION TRUST) danyal.khan@nhs.net
Cc: MARCUS, Hani (UNIVERSITY COLLEGE LONDON HOSPITALS NHS FOUNDATION TRUST) hani.marcus@nhs.net

Good morning Danyal,

Hope you are well.

I am the costing officer reviewing the cost of this study.

I have completed my review of the SoECAT on the CPMS, I am happy for this to be approved.

Hi Suzanne, I am happy for you to proceed to sign off.

Please let me know if you have any questions.

Kind Regards,
Michelle Gyekye
Cost Accountant, UCLH

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4th Floor, West
250 Euston Road
London
NW1 2PG
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Working hours: 08:00 to 16:00. Monday – Friday

From: SWAIN, Suzanne (UNIVERSITY COLLEGE LONDON HOSPITALS NHS FOUNDATION TRUST) <suzanne.swain1@nhs.net>
Sent: 16 January 2024 17:19
To: RD.FINANCE(0CC61) (UNIVERSITY COLLEGE LONDON HOSPITALS NHS FOUNDATION TRUST) <uclh.enquiry.rd.finance@nhs.net>
Subject: FW: EDGE ID 163765- NIHR PhD Fellowship Application - SoECAT query

Dear Vince

Please can you assign for a Grant application costing review

Suzanne Swain
Sponsorship Officer

Please note my working days are Tuesday, Wednesday, Thursday.

UCLH/UCL Joint Research Office,
part of the Research Directorate
4th Floor, West, 250 Euston Road, London, NW1 2PG
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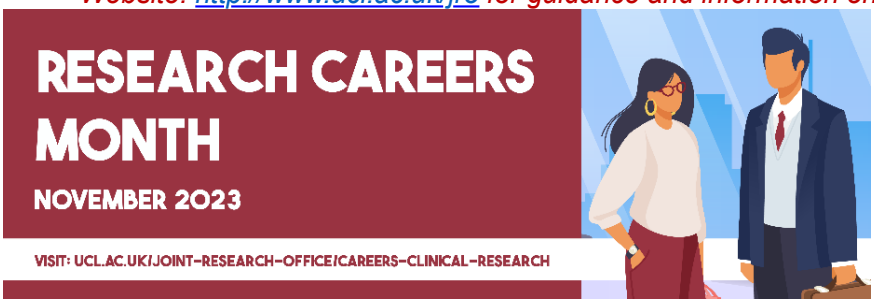
Website: http://www.ucl.ac.uk/jro for guidance and information on research at UCL/UCLH.



From: RANDD (UNIVERSITY COLLEGE LONDON HOSPITALS NHS FOUNDATION TRUST) <uclh.randd@nhs.net>
Sent: Thursday, January 11, 2024 3:09 PM
To: KHAN, Danyal (UNIVERSITY COLLEGE LONDON HOSPITALS NHS FOUNDATION TRUST) <danyal.khan@nhs.net>
Cc: MARCUS, Hani (UNIVERSITY COLLEGE LONDON HOSPITALS NHS FOUNDATION TRUST) <hani.marcus@nhs.net>; SWAIN, Suzanne (UNIVERSITY COLLEGE LONDON HOSPITALS NHS FOUNDATION TRUST) <suzanne.swain1@nhs.net>
Subject: Re: EDGE ID 163765- NIHR PhD Fellowship Application - SoECAT query

Dear Danyal and team,

Re: Local project ref: 163765
Study title: Towards AI-Assisted Pituitary Surgery – The Systematic Translation of Surgical Workflow Analysis to Improve Training, Audit and Prognostication.

Re: Grant Application Registered for Review and Costing

I acknowledge receipt of your grant application. I can confirm that your submission is valid and has been registered.

Your application has been assigned to Suzanne who will contact you regarding next steps.

Kind Regards

Rabia Bibi
Portfolio and Database Administrator, UCLH

UCLH/UCL Joint Research Office (part of the Research Directorate)
4th Floor, West, 250 Euston Road, London, NW1 2PG
Web: www.ucl.ac.uk/jro

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From: "KHAN, Danyal (UNIVERSITY COLLEGE LONDON HOSPITALS NHS FOUNDATION TRUST)" <danyal.khan@nhs.net>
Date: Thursday, 11 January 2024 at 15:01
To: "RANDD (UNIVERSITY COLLEGE LONDON HOSPITALS NHS FOUNDATION TRUST)" <uclh.randd@nhs.net>
Cc: "MARCUS, Hani (UNIVERSITY COLLEGE LONDON HOSPITALS NHS FOUNDATION TRUST)" <hani.marcus@nhs.net>
Subject: Re: EDGE ID 163765- NIHR PhD Fellowship Application - SoECAT query

Thank you, please find attached

BW
Danyal

> On 11 Jan 2024, at 13:51, RANDD (UNIVERSITY COLLEGE LONDON HOSPITALS NHS FOUNDATION TRUST) <uclh.randd@nhs.net> wrote:
>
> Dear Danyal,
>
> Please provide completed grant form
>
> Kind Regards
> ---
>
> **Rabia Bibi**
> **Portfolio and Database Administrator, UCLH**
> UCLH/UCL Joint Research Office (part of the Research Directorate)
> 4th Floor, West, 250 Euston Road, London, NW1 2PG
> Web: www.ucl.ac.uk/jro
>
> We are committed to delivering top-quality patient care, excellent education and world class research
> **safety kindness teamwork Improving**

<image001.png>

From: "KHAN, Danyal (UNIVERSITY COLLEGE LONDON HOSPITALS NHS FOUNDATION TRUST)" <danyal.khan@nhs.net>
Date: Thursday, 11 January 2024 at 13:43
To: "RANDD (UNIVERSITY COLLEGE LONDON HOSPITALS NHS FOUNDATION TRUST)" <uclh.randd@nhs.net>
Cc: "MARCUS, Hani (UNIVERSITY COLLEGE LONDON HOSPITALS NHS FOUNDATION TRUST)" <hani.marcus@nhs.net>
Subject: Re: EDGE ID 163765- NIHR PhD Fellowship Application - SoECAT query

Hi Rabia

Ive already done an online SoECAT on CPMS, the CPMS ID is 58564, as per the attached

Ive attached the draft application, and there is no funding letter yet as Im still only in the application stage

Best wishes,
Dan

Danyal Khan
Neurosurgery Registrar
Royal London Hospital

> On 11 Jan 2024, at 13:39, RANDD (UNIVERSITY COLLEGE LONDON HOSPITALS NHS FOUNDATION TRUST) <uclh.randd@nhs.net> wrote:
>
> Dear Danyal,
>
> **Re: LRN 163765- Grant Application submission**
>
> In order that the application can be registered and assigned for review, please can you submit relevant documents as listed below, to validate and register the application.
>
> 1. Online SoECAT Application – please see below guidance
> 2. Protocol and/or grant application
> 3. Funding letter(s) (if grant secured)
> 4. Copy of the draft grant application form (attached)
>
> **Guidance for the completion of the SoECAT and creation of a CPMS account (to fill in the Online SoECAT) by the applicant is present in the online tool to assist at each page and stage of the application**

process and further details can be found on the Online SoECAT Guidance page here

There is also an Online SoECAT Guidance Module which includes video tutorials and linked resources (an NIHR Learn account is required to access and enrol onto the module) which you can access here

On receipt of a valid submission, the study will be registered and assigned to a Sponsorship Officer who will contact you regarding next step. I will confirm with you the name of the Officer when the study has been registered.

Kind Regards

Rabia Bibi
Portfolio and Database Administrator, UCLH
UCLH/UCL Joint Research Office (part of the Research Directorate)
4th Floor, West, 250 Euston Road, London, NW1 2PG
Web: www.ucl.ac.uk/jro

We are committed to delivering top-quality patient care, excellent education and world class research
safety kindness teamwork Improving

<image001.png>

From: "KHAN, Danyal (UNIVERSITY COLLEGE LONDON HOSPITALS NHS FOUNDATION TRUST)" <danyal.khan@nhs.net>
Date: Thursday, 11 January 2024 at 13:27
To: "RANDD (UNIVERSITY COLLEGE LONDON HOSPITALS NHS FOUNDATION TRUST)" <uclh.randd@nhs.net>
Cc: "MARCUS, Hani (UNIVERSITY COLLEGE LONDON HOSPITALS NHS FOUNDATION TRUST)" <hani.marcus@nhs.net>
Subject: NIHR PhD Fellowship Application - SoECAT query

Dear JRO team

I am a neurosurgery ACF, applying for an NIHR Doctoral Fellowship. Please see my attached fellowship application which, in short , seeks to perform single-centre proof of concept and pilot studies of AI interventions at NHNN.

I have an approved SoECAT (see attached "Funder Export" doc) that I worked on with the ACORD team at NIHR. It was based on the approved SoECAT for the relevant ethics for my fellowship studies which is part of the JRO portfolio

my fellowship studies which is part of the CRTG portfolio already (E127474 - see attached)

In the application, it asks whether the costs have been discussed with the trust. It is a low-cost study, with the only real “costs” being small amounts of time for staff to complete surveys. I trust this updated "Funder Export” SoECAT looks ok to you too?

Best wishes,
Dan

Danyal Khan
Neurosurgery Registrar | Royal London Hospital
NIHR Academic Clinical Fellow | University College London
@dzkhan94
<Grant Form.docx>

<Grant Form.docx>

Grant Form_...1].docx	NIHR Doctoral applic.a[2].docx	Funder Export...[2].xlsx 17 KB	E127474[1].xlsx 81 KB

From: **No-Reply CPMS SoECAT** no-reply-CPMS-SoECAT@nihr.ac.uk
Subject: SoECAT Authorised: CPMS 58564, IRAS 127474, AI-assisted Pituitary Surgery
Date: 9 November 2023 at 10:11
To: dzkhan94@gmail.com

NIHR | National Institute for Health Research

Appendix #5: Funder Export-127474_SoECAT.xlsx